



October 6, 2023

Office of Food Additive Safety (HFS-200)
Center for Food Safety and Applied Nutrition
Food and Drug Administration
5001 Campus Drive
College Park, MD 20740-3835

Subject: GRAS Notification for Use of Fava Bean Protein in Food

Dear Sir:

On behalf of Nuritas, ToxStrategies, Inc. (its agent) is submitting, for FDA review, a copy of the GRAS Notification as required. The enclosed document provides notice of a claim that the food ingredient, fava bean protein, described in the enclosed notification is exempt from the premarket approval requirement of the Federal Food, Drug, and Cosmetic Act because it has been determined to be generally recognized as safe (GRAS), based on scientific procedures, for addition to food.

If you have any questions or require additional information, please do not hesitate to contact me at 630-352-0303, or dschmitt@toxstrategies.com.

Sincerely,

Donald F. Schmitt, M.P.H.
Senior Managing Scientist

GRAS Determination of Fava Bean Protein Hydrolysate for Use as an Ingredient in Human Food

SEPTEMBER 22, 2023

ToxStrategies

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GRAS Determination of Fava Bean Protein Hydrolysate for Use as an Ingredient in Human Food

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List of Acronyms

ALT	alanine aminotransferase
AST	aspartate aminotransferase
bw	body weight
cGMP	current Good Manufacturing Practice
C	centigrade
CAS	Chemical Abstracts Service
CFR	Code of Federal Regulations
CFU	colony-forming units
CK	creatine kinase
COA	Certificate of Analysis
db	dry basis
FFDCA	Federal Food, Drug, and Cosmetic Act
FDA	U.S. Food and Drug Administration
FFA	free fatty acids
g	gram
G6PD	glucose-6-phosphate dehydrogenase
GGT	gamma-glutamyl transferase
GRAS	Generally Recognized as Safe
GRN	Generally Recognized as Safe Notification
HMG	3-hydroxy-3-methylglutaryl
Hs-CRP	high-sensitivity C-reactive protein
IgE	immunoglobulin E
IOM	Institute of Medicine
JECFA	Joint FAO/WHO Expert Committee on Food Additives
kcal	kilocalories
kg	kilogram
LD ₅₀	lethal dose in 50 percent of treated animals
LDH	lactate dehydrogenase
LDL	low-density lipoprotein
mg	milligram
MPC	milk protein concentrate
NOEL	no-observed-effect level
RBC	red blood cell
RDA	Recommended Dietary Allowance
RTD	ready to drink
TG	triglycerides
US	United States
USC	United States Code
USDA	United States Department of Agriculture
V/C	vicine/convicine
VLDL	very low-density lipoprotein
WHO	World Health Organization

§ 170.225 Part 1, GRAS Notice: Signed Statements and Certification

(1) GRAS Notice Submission

Nuritas Ltd., through its agent, ToxStrategies, Inc., hereby notifies the U.S. Food and Drug Administration (FDA) of the submission of a Generally Recognized as Safe (GRAS) notice for the use of fava bean protein hydrolysate in select foods for human consumption in accordance with Subpart E of 21 CFR § 170.

(2) Name and Address

Nuritas
19B Joshua Dawson House
Dawson St
Dublin 2
Ireland

(3) Name of Notified Substance

The name of the substance that is the subject of this GRAS determination is a fava bean protein hydrolysate from *Vicia faba* L., also referred to as broad bean, horse bean, field bean, and faba bean.

(4) Intended Use in Food

The fava bean protein hydrolysate is proposed for use only as an alternative to other plant-based protein ingredients in select foods for human consumption (except for infant formula), and the daily consumption of protein is not expected to increase as a result of its introduction.

(5) Statutory Basis for GRAS Determination

Nuritas, through its agent, ToxStrategies, confirms that the fava bean protein hydrolysate ingredient, which meets the specifications described herein, has been determined to be GRAS through scientific procedures in accordance with 21 CFR § 570.30(a) and (b).

(6) Premarket Approval Statement

Nuritas further asserts that the use of the fava bean protein hydrolysate ingredient, as described herein in specified foods, is exempt from the pre-market approval requirements of the Federal Food, Drug, and Cosmetic Act, based on a conclusion that the substance is GRAS under the conditions of its intended use.

(7) Availability of Information

Should this GRAS determination be submitted to the FDA, the data and information that serve as the basis for this GRAS determination, as well as any information that has become available since the GRAS determination, will be sent on request, or are available for the FDA's review and copying during customary business hours from ToxStrategies, Inc., Naperville, IL.

(8) Data and Information Confidentiality Statement

None of the data and information in the GRAS notice are exempt from disclosure under the Freedom of Information Act, 5 U.S.C. 552.

(9) GRAS Certification

To the best of our knowledge, this GRAS determination is a complete, representative, and balanced document. Nuritas is not aware of any information that would be inconsistent with a finding that the proposed uses and use levels of the fava bean protein hydrolysate ingredient in specified foods, meeting the appropriate specifications described herein, and used according to current Good Manufacturing Practice (cGMP), is GRAS. Recent reviews of the scientific literature revealed no potential adverse health concerns.

(10) Name/Position of Notifier


Donald F. Schmitt, M.P.H.
Senior Managing Scientist
ToxStrategies, Inc.
Agent for Nuritas

10/06/2023
Date

(11) FSIS Statement

The fava bean protein hydrolysate ingredient will not be used in products under the jurisdiction of the U.S. Department of Agriculture (USDA).

§ 170.230 Part 2, Identity, Method of Manufacture, Specifications, and Physical or Technical Effect

A. Identity

The fava bean protein hydrolysate is a hydrolysis product of fava beans consisting of approximately 55-65 % protein. It is a beige/off yellow powder with no off-odor or flavor.

B. Common or Usual Name

Fava bean protein hydrolysate. The ingredient will be referred to as fava bean protein hydrolysate throughout the document.

C. CAS Registry Number

Not applicable.

D. Trade Name

The trade names for the fava bean protein hydrolysate are PeptiStrong™, PeptiNourish™, PeptiTone™, and 4X protein™.

E. Fava Bean Protein Hydrolysate Composition

Typical nutritional data for the fava bean protein hydrolysate is summarized in Table 1. The carbohydrate portion of the fava bean protein hydrolysate is primarily sugars such as glucose, fructose, sucrose, raffinose, stachyose, and verbascose (Landry et al., 2016). Table 2 presents the alkaloid concentrations (anti-nutrient components vicine, convicine) in three lots of the proposed fava bean protein hydrolysate product. Levodopa, also known as L-DOPA or L-3,4-dihydroxyphenylalanine, naturally occurs in fava beans, but its levels decrease significantly in heated/cooked beans (Carador-Martinez et al., 2012; Multari et al, 2015). The level of L-DOPA in the fava bean protein isolate was determined to be approximately 1400 mg/kg.

The fava bean protein hydrolysate, due to its high protein content, is also rich in amino acids and Table 3 presents a comparison of its amino acid profile to other common food-derived proteins.

Table 1. Typical nutritional data for fava bean protein hydrolysate

Protein (%)	55-65%
Moisture (%)	3-8%
Carbohydrates (%)	25-35%
Ash (%)	5-6%

Fat (%)	1-4%
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Table 2. Alkaloid concentration in three lots of fava bean protein hydrolysate (mg/g) *

Alkaloid	Batch No. sh_CU6G3M	Batch No. sh_YO75MA	Batch No. sh_FDE4MQ
Vicine	14.1	11.3	0.54
Convicine	4.7	4.1	0.155
Sum	18.8	15.4	0.695

*by inverse phase UPLC method

Table 3. Typical amino acid profile (% dry basis)¹

Amino Acid	Fava Bean Protein Hydrolysate*	Fava Bean Protein; GRN 879	Oat	Wheat	Rice	Soy	Whey
Alanine	2.18	4.06	4.37	4.0	5.67	NA	NA
Arginine	5.23	8.58	7.17	2.7	7.64	NA	NA
Aspartic Acid	6.08	9.90	7.48	3.2	9.33	NA	NA
Cysteine	0.55	0.87	2.45	1.3	2.09	NA	NA
Glutamic Acid	9.06	16.8	22.5	33.7	17.22	NA	NA
Glycine	2.25	3.67	4.13	5.1	4.33	NA	NA
Histidine	1.41	2.37	2.22	2.14	2.12	2.3	2.2
Isoleucine	1.88	4.40	4.38	3.39	4.4	4.7	5.8
Leucine	3.78	8.05	8.42	6.67	8.3	6.6	12
Lysine	3.59	6.15	3.38	2.5	3.4	5.4	10.8
Methionine	0.41	0.79	2.23	1.4	2.66	NA	NA
Methionine + Cysteine	0.96	1.66	4.68	4.33	5.61	2.9	4.2
Phenylalanine	2.23	4.51	5.95	4.2	5.38	NA	NA
Phenylalanine + Tyrosine	4.05	7.90	10	7.6	10.85	9.8	5.1
Proline	2.48	4.27	5.57	14.2	4.49	NA	NA
Serine	3.07	4.81	3.98	5.9	4.75	NA	NA
Threonine	2.02	3.33	3.5	2.96	3.7	4	7.2
Tryptophan	NA	1.03	1.1	1.21	1.2	1.2	2.1
Tyrosine	1.82	3.39	4.05	2.7	4.84	NA	NA
Valine	2.08	4.85	5.78	4.42	6	4.2	5.8

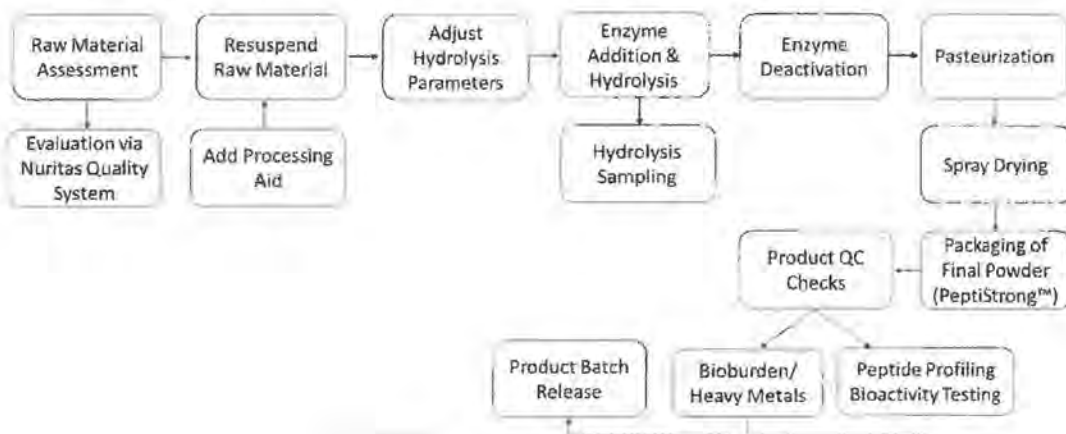
¹ Amino acid values for the other food proteins were excerpted from GRAS Notification (GRN) No. 575 for oat protein, GRN Nos. 26 and 575 for wheat protein, GRN Nos. 609 and 575 for rice protein, GRN No. 575 for soy protein, and GRN No. 575 for whey protein. NA = not available; * = average of three lots

F. Manufacturing Process

A diagram of the fava bean protein product manufacturing process is shown below (Figure 1).

The fava bean protein hydrolysate ingredient (approximately 60% protein) is derived from a fava bean protein starting material in a process that involves a series of steps with common, safe, and suitable processing aids (all have U.S. regulatory approval) and without the use of organic solvents (see Table 4). The protein powder starting material is resuspended in a suitable food grade alkalizing buffer and heated. Following pH adjustment, if required, with an acid suitable for food production, a commercially available, microbial-derived protease enzyme is added to catalyze the hydrolysis reaction. This enzyme preparation is GRAS in accordance with 21 CFR § 184.1(b)(1), and approved for use in food with no limitation other than current good manufacturing practice. The affirmation of the use this enzyme in food is based upon the following current good manufacturing practice conditions of use: (1) The ingredient is used as an enzyme as defined in § 170.3(o)(9) of this chapter to hydrolyze proteins or polypeptides. (2) The ingredient is used in food at levels not to exceed current good manufacturing practice¹. After a defined period, the reaction is deactivated with a thermal heat treatment step before spray-drying. These steps ensure that no residual enzyme activity remains in the final product.

Figure 1. Steps in the fava bean protein hydrolysate production process



¹ [https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-184#p-184.1150\(c\)](https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-184#p-184.1150(c))

Table 4. Processing aids

Processing Aid	CAS Number(s)	21 CFR/GRN Citation(s)
Ammonium bicarbonate (Alternatively, also other alkaline hydroxides permitted for such use)	1066-33-7	21 CFR §184.1135
Sulfuric acid (Alternatively, also other acids permitted for such use)	7664-93-9	21 CFR §184.1095
Endoproteases/endopeptidases	Listed in internal classification systems, (e.g., EC 3.4.21.62, which hydrolyzed internal protein bonds)	21 CFR §184.1150

G. Characterization of *Vicia faba* L.

Fava bean (*Vicia faba* L.), also referred to as broad bean, horse bean, faba bean, and field bean, is an early legume crop and belongs to the *Fabaceae* family. There is little evidence of the origins of its domestication, as its wild progenitor is still unknown. However, the oldest seeds of fava bean were found in the late 10th millennium, in north-west Syria (Willcox and Tanno, 2006). The fava bean has a long history of use as feed and food, likely because of its valuable content of both protein and energy (Crepon et al., 2010). The fava bean is a rich source of fiber and non-nutrient secondary with a protein content about twice that of cereals and several times that of root tubers. Fava beans are a rich source of proteins, carbohydrates, fiber, vitamins, and minerals (Aune et al., 2011).

H. Product Specifications

Specifications for the product are presented in Table 5. Analytical results from three lots are provided in Appendix A. A comparison of the three lots of the product to the specifications below can be found in Table 6.

Table 5. Specifications for fava bean protein hydrolysate

Parameter	Specification	Method
Identity		
Appearance	Powder	
Color	Light beige	Visual
Odor	Natural with no off odor	--
Flavor	Natural with no off flavor	--
Protein (% db)	55-65	AOAC 992.33
Physical and Chemical Parameters		
Moisture (%)	3-8	AOAC 927.05

Heavy Metals		
Lead (mg/kg)	≤0.05	ICP-MS
Arsenic (mg/kg)	≤0.05	ICP-MS
Cadmium (mg/kg)	≤0.05	ICP-MS
Mercury (mg/kg)	≤0.05	ICP-MS
Microbiological Parameters		
Total Plate Count (CFU/g)	≤10,000	AOAC 990.12
Yeast and mold (CFU/g)	≤1000	AOAC 997.02
E. coli (CFU/g)	≤20	AOAC 991.14
Coliforms (CFU/g)	≤20	AOAC 991.14
Salmonella (CFU/25g)	Negative	AOAC 2013.02

Table 6. Analytical results for three lots of fava bean protein hydrolysate

Specification		Batch No. sh_CU6G3M	Batch No. sh_YO75MA	Batch No. sh_FDE4MQ
Protein (%)	55-65	58.98	56.45	60
Moisture (%)	3-8	5.26	6.31	4.87
Lead (mg/kg)	≤0.05	<0.01	<0.01	<0.007
Arsenic (mg/kg)	≤0.05	<0.01	<0.01	0.014
Cadmium (mg/kg)	≤0.05	0.015	0.011	0.015
Mercury (mg/kg)	≤0.05	<0.01	<0.01	<0.013
Total Plate Count (CFU/g)	≤10,000	300	2,300	240
Yeast and mold (CFU/g)	≤1000	<10	<100	<10
E. coli (CFU/g)	≤20	<10	<10	<10
Coliforms (CFU/g)	≤20	<10	<10	<10
Salmonella (CFU/375g x 2)	Negative	Negative	Negative	Negative

The analytical (physical, chemical, and microbiological) results for the fava bean protein hydrolysate ingredient summarized in the above tables and included in the Certificate of Analyses in Appendix A confirm that the ingredient meets the proposed analytical specifications and demonstrates the consistency of production. Additional analytical results included in Appendix A also confirm the lack of other impurities/contaminants (e.g., pesticides, aflatoxins).

I. Stability Data

The stability of fava bean protein hydrolysate has been tested under normal conditions (25°C and 60% relative humidity) for 24 months. Technical specifications for the product include shelf-life storage conditions of 24 months from the date of manufacture when stored in a closed container in a cool (below 100°F) and dry place away from strong light. Stability testing data can be found in Table 7.

Table 7. Stability testing data for fava bean protein hydrolysate

Specifications			Stability Intervals (months)						
Test	Acceptance Criteria	Units	0	3	6	9	12	18	24
Organoleptic Properties:									
Taste	Palatable	n/a	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms
Color	Beige, off yellow	n/a	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms
Odor	Bland, no off-odor	n/a	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms
Flavor	Bland, no off-flavor	n/a	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms	Conforms
Microbiological Properties:									
Aerobic colony count	< 10 ⁴	CFU/g	300	60	10	40	30	<10	<10
Bacillus cereus (Presumptive)	< 10 ³	CFU/g	100	200	100	200	100	100	<100
E. coli	< 20	CFU/g	< 10	<10	<10	<10	<10	<10	<10
Salmonella	Not Detected in 25g	CFU/25g	Not Detected	Not Detected	Not Detected	Not Detected	Not Detected	Not Detected	Not Detected
Staphylococci (& Coagulase)	< 20	CFU/25g	Absent	<20	<20	<20	<20	<20	<20
Yeasts & Moulds	< 100	CFU/g	< 10	<100	<100	<100	<100	<100	<100
Physicochemical Properties:									
pH	Range pH 5-7	pH	6.08	6.12	6.24	6.12	6.08	6.06	6.11
Protein Content	55-65	%	59.4	58.7	58.6	57.2	57.9	60	57
Water activity	< 0.8	Aw	n/a	0.472	0.399	0.29	0.438	0.484	0.395
Moisture Content	< 10%	%	5.26	5.4	4.1	5.53	4.92	5.83	5.77

*Lot No. CU6G3M

§ 170.235 Part 3, Dietary Exposure

Proposed Use

The focus of this GRAS determination of fava bean protein hydrolysate is for use in certain specified conventional foods at levels similar to those that have been recognized in previous GRNs for current plant-based protein sources such as soy (GRN No. 134; FDA 2004), canola (GRN Nos. 327, 386, 683; FDA 2010, 2011, 2017a), pea (GRN Nos. 182, 581, 608, 788; FDA 2005, 2015b, 2016a, 2018b), wheat (GRN Nos. 26 and 182; FDA 1999, 2005), rice (GRN No. 609, 944; FDA 2016b, 2020b), whey (GRN Nos. 37, 633; FDA 2000, 2016c), potato (GRN No. 447; FDA 2013), oat (GRN 575; FDA 2015a), mung bean (GRN 684; FDA 2017b), hemp seed (GRN 771; FDA 2018a) fava bean protein (GRN 879; FDA 2020a) and rice protein hydrolysate (GRN 944; 2020b). Similarly, fava bean–derived protein hydrolysate is intended for use as a source of protein for enrichment of certain specified processed foods listed below.

The proposed uses of the fava bean protein hydrolysate ingredient for enrichment of foods include the following food categories (see Table below):

- Dry blend powders - nutritionally fortified beverage mixes
- Meal replacement drinks/meal replacement bars
- Dairy alternatives, dairy analogs
- Plant-based burgers
- Ready to drink (RTD) beverages - protein shakes, smoothies, sports drinks, etc.
- Nutritionally fortified confectionary – hard candies, soft gummies etc.
- Nutritionally fortified snack foods – protein bars, energy bars etc.
- Baked goods – nutritionally fortified cookies, biscuits, cereal bars
- Meat alternatives-such as vegan burger, etc.

The fava bean protein hydrolysate ingredient will be added at a maximum of 2.4 g/serving or given a protein content of approximately 60%, approximately 1.44 g protein/serving.

Table 8. Food categories and use levels

Food Category	Food Subcategory	Maximum Proposed Use Level (g/serving)
Dry Blend Powders	Protein Powders	2.4g
	Instant Breakfast	2.4g
	Sports Drink Blends	2.4g
Meal replacements	Drinks	2.4g
	Bars	2.4g
Dairy Alternatives	Milk Analogs	2.4g
	Grits	2.4g

	Oatmeal	2.4g
RTD beverages	Protein Shakes	2.4g
	Sports Drinks	2.4g
Nutritionally fortified confectionary	Hard Candies	2.4g
	Gummies	2.4g
Nutritionally fortified snack foods	Protein Bars	2.4g
	Energy Bars	2.4g
Baked Goods	Cookies	2.4g
	Biscuits	2.4g
	Cereal Bars	2.4g
Meat Alternatives	Veggie Burger	2.4g
	Meatless Fish Stick	2.4g
	Meatless Pork/Beef	2.4g

The Institute of Medicine (IOM, 2005) recommends that adults consume 0.8 grams of protein per kilogram of body weight. IOM also set a wide range for acceptable protein intake, ranging from 10 - 35% of calories each day. In the U.S., the recommended daily allowance (RDA) of protein is 46 grams/day for women over 19 years of age, and 56 grams/day for men over 19 years of age.”

However, the RDA does not represent a safety based upper limit of protein consumption. Physically active persons on normal diets are known to exceed this level, and individuals involved in bodybuilding ingest much higher levels of protein (WHO, 2002). The accepted WHO safe level of intake is 0.83 g/kg per day, for proteins with a protein digestibility-corrected amino acid score value of 1.0. While WHO has stated that no safe upper limit has been identified, they also indicated that it is unlikely that intakes of twice the safe level are associated with any risk to healthy individuals.

Dietary Survey Data

Dietary survey data were obtained from What We Eat in America (WWEIA), the dietary interview portion of the National Health and Nutrition Examination Survey (NHANES). NHANES is carried out in 2-year cycles (biennials) by the Centers for Disease Control and Prevention (CDC), to characterize the general health and nutritional status of children and adults across the U.S. The dietary intake survey portion of NHANES is a joint effort between CDC and the U.S. Department of Agriculture (USDA). All NHANES biennials for which dietary intake data were available were included in this analysis (2003–2004, 2005–2006, 2007–2008, 2009–2010, 2011–2012, 2013–2014, and 2015–2016).

The first day of the WWEIA dietary questionnaire was administered in person, in conjunction with the participants' interviews and examinations for the other NHANES lifestyle and laboratory assessments. The second day of the survey was collected via a phone interview at some point 3–10 days after the first survey day. Data collected during the dietary interview include foods consumed by the participant, encoded by a food code, and amounts eaten. Respondents who provided complete records for both days were designated as reliable by WWEIA; only reliable respondents who also provided body-weight data were considered in this analysis.

Methods

To estimate the intake of fava bean protein hydrolysate from its proposed uses in specified foods, ToxStrategies performed the following steps:

- Step 1: Identified foods and their components to which fava bean protein hydrolysate may be added
- Step 2: Calculated individual intake of fava bean protein hydrolysate for individual survey participants
- Step 3: Calculated population statistics estimating fava bean protein hydrolysate intake
- Step 4: Estimate protein contribution of fava bean protein hydrolysate addition.

Identification of Foods and Their Components to Which Fava Bean Protein Hydrolysate May Be Added

To identify foods that are proposed to contain the fava bean protein hydrolysate, ToxStrategies performed a thorough search of food codes reported in WWEIA. Food code descriptions from WWEIA were examined manually and flagged if they could be considered members of one or more of the food categories specified above. Food codes retained for further analysis can be found in Appendix B.

Calculation of Individual Intake of Fava Bean Protein Hydrolysate for Individual Survey Participants

Only those respondents designated as reliable were included in this assessment.

Both days of the NHANES WWEIA dietary interviews from each biennial were analyzed. Participants' consumption of the fava bean protein hydrolysate was averaged over the two response days, *i.e.*, $(\text{Day1 consumption} + \text{Day2 consumption})/2$. Consumption of fava bean protein hydrolysate was calculated using the grams of the relevant food consumed, as reported in NHANES, multiplied by the maximum proposed use level (2.4 g/serving) of fava bean protein hydrolysate in that food. For example, for the food "53720500, Snickers Marathon Protein Bar", the total food weight is 80 g per

serving. Thus, a survey participant who consumed this food would consume 0.03 g of fava bean protein hydrolysate per gram of the food.

Calculation of Population Statistics Describing Fava Bean Protein Hydrolysate Estimated Daily Intake

All NHANES biennials for which dietary intake data was available were included in this analysis (2003-2004, 2005-2006, 2007-2008, 2009-2010, 2011-2012, 2013-2014, and 2015-2016). The dietary and sample weighting data from the two biennials were combined according to the NHANES analytic guidelines for combining surveys. From the combined dataset, we estimated survey-design-weighted descriptive statistics for the population consumption per day.

Population statistics were estimated using the “survey” package version 3.36 (Lumley, 2004) in the R (version 3.6.1) environment for statistical computing (R Core Team, 2019) using the appropriate adjustment to sampling weights for combining biennials, then incorporating survey sampling units and strata from the survey design to ensure that sub-populations and areas were correctly represented. Descriptive statistics (mean, 90th percentile) were calculated for the subset of consumers of fava bean protein hydrolysate and for the entire population; they were broken down by age range and body weight adjustment. Values were provided by individual food category and for total consumption of foods in all food categories.

The tables below present the EDI for fava bean protein hydrolysate and protein in grams per day and grams per kilogram body weight per day for the following age groups in the US population: 2 years and older, 2 to 5 years, 6 to 18 years, 0 to 80 years, and 19 years and older. The “number of users” refers to the number of survey participants in a given age group who consumed a food item in given food category. The “percent users” is the percentage of fava bean protein hydrolysate users out of the total number of reliable survey participants (users and non-users) belonging to a given age group.

Table 9. Estimated daily intake for fava bean protein hydrolysate (g/day)

Age Group	Number of users	Percent users	EDI per user (g/day)		EDI per capita (g/day)	
			Mean	90th percentile	Mean	90th percentile
2-80	6866	12%	6.44	12.4	0.922	2.88
2-5	477	9%	4.21	7.17	0.439	0.586
6-18	2066	13%	5.64	11.2	0.860	2.88
0-80	7041	12%	6.44	12.4	0.906	2.88
19-80	4323	12%	6.76	13.3	0.971	2.88

Table 10. Estimated daily intake for fava bean protein hydrolysate (g/kg-bw/day)

Age Group	Number of users	Percent users	EDI per user (g/kg-bw/day)		EDI per capita (g/kg-bw/day)	
			Mean	90th percentile	Mean	90th percentile
2-80	6866	12%	0.099	0.199	0.0142	0.0416
2-5	477	9%	0.258	0.436	0.0269	0.0320
6-18	2066	13%	0.116	0.217	0.0177	0.0592
0-80	7041	12%	0.104	0.205	0.0147	0.0410
19-80	4323	12%	0.0865	0.172	0.0124	0.0389

Table 11. Estimated daily intake for fava bean protein (g/day)

Age Group	Number of users	Percent users	EDI per user (g/day)		EDI per capita (g/day)	
			Mean	90th percentile	Mean	90th percentile
2-80	6866	12%	3.86	7.45	0.553	1.73
2-5	477	9%	2.52	4.30	0.263	0.351
6-18	2066	13%	3.38	6.70	0.516	1.73
0-80	7041	12%	3.86	7.45	0.544	1.73
19-80	4323	12%	4.06	7.99	0.583	1.73

Table 12. Estimated daily intake for fava bean protein (g/kg-bw/day)

Age Group	Number of users	Percent users	EDI per user (g/kg-bw/day)		EDI per capita (g/kg-bw/day)	
			Mean	90th percentile	Mean	90th percentile
2-80	6866	12%	0.0594	0.120	0.00851	0.0250
2-5	477	9%	0.155	0.262	0.0162	0.0192
6-18	2066	13%	0.0696	0.130	0.0106	0.0355
0-80	7041	12%	0.0625	0.123	0.00880	0.0246
19-80	4323	12%	0.0519	0.103	0.00746	0.0234

Nuritas' proposed fava bean protein hydrolysate ingredient is intended only to be an alternative source of protein for the above uses in specified foods. As was concluded in the other GRAS notifications, we do not realistically expect that the actual consumption of foods containing fava bean-derived protein hydrolysate would result in daily consumption greater than the RDA for protein. Most of the population's intake of protein is, and will remain, in the form of unprocessed foods, including meat, poultry, fish, and legumes. The proposed fava bean protein hydrolysate product is but one of many protein sources for use in processed foods, so only the known conservatism of intake calculations such as those described in the aforementioned GRNs suggest any possibility of exceeding the RDA at the 90th percentile (FDA, 2010, 2011).

In summary, the proposed uses of fava bean-derived protein hydrolysate in select specified foods will not result in an increase in the overall consumption of protein, but simply provide an alternative source of well-characterized protein hydrolysate from fava beans for use in specified foods.

§ 170.240 Part 4, Self-Limiting Levels of Use

The use of fava bean protein hydrolysate in protein-enriched foods is considered self-limiting for technological reasons, such as product texture and/or flavor profile.

§ 170.245 Part 5, Experience Based on Common Use in Food

While there exists broad historical evidence of fava beans and the protein contained therein as food for human consumption, the statutory basis for our conclusion of the GRAS status of the fava bean protein hydrolysate in this notice is based on scientific procedures and not common use in food.

§ 170.250 Part 6, GRAS Narrative

History of Use and Regulatory Approval

FDA has reviewed published information and data on many protein products as part of GRAS notifications for animal and plant-based protein isolates, concentrates and hydrolysates and subsequently issued “no objection letters.” The previous GRNs for plant-based protein sources include soy (GRN No. 134; FDA 2004), canola (GRN Nos. 327, 386, 683; FDA 2010, 2011, 2017a), pea (GRN Nos. 182, 581, 608, 788; FDA 2005, 2015b, 2016a, 2018b), wheat (GRN Nos. 26 and 182; FDA 1999, 2005), rice (GRN No. 609, 944; FDA 2016b, 2020b), whey (GRN Nos. 37, 633; FDA 2000, 2016c), potato (GRN No. 447; FDA 2013), oat (GRN 575; FDA 2015a), mung bean (GRN 684; FDA 2017b), hemp seed (GRN 771; FDA 2018a), fava bean protein (GRN 879; FDA 2020a), and rice protein hydrolysate (GRN 944: 2020b).

GRN 879 (FDA, 2020a) for the use of fava bean protein in human foods in the United States received a letter of no objection from FDA in 2020. It should be noted that there are references in the public domain (e.g., general interest) to fava bean protein ingredients being self-determined as GRAS (e.g., Ingredion Vitessence™, approximately 60% protein). In addition, fava beans and other legumes, and the protein from these sources, have been commonly consumed as food and feed around the world for decades (Crepon et al., 2010; Schmandke et al., 2010).

Safety

Introduction

Literature searches were performed to identify available safety data on fava beans, fava bean protein, and the associated anti-nutrients (through July 2022) reported in them. This included searching sources of information such as publicly available assessments, databases, or reviews from organizations, including EFSA, Joint FAO/WHO Expert Committee on Food Additives (JECFA), U.S. FDA, and the World Health Organization (WHO), general internet searching, and searching databases such as Embase, MedLine, ToxLine, and PubMed.

As described previously, the fava bean can be used throughout the year and is consumed in both raw and processed forms. In the human diet, it is mostly the seed grain that is consumed, while the pods are used as animal feed. Fava beans contain approximately 250 g protein/kg seed and provides 320 kcal/100 g dry weight of energy. Fava bean, its fractions, and its processing products (grains, hulls, and flours) also contain anti-nutritional factors, however, typical soaking and heat processing steps including cooking, boiling, autoclaving, or extrusion cooking steps can reduce the amounts of the anti-nutritional factors by up to 100%, thus limiting any safety concerns (Jamalian et al., 1999).

Fava bean is also a rich source of fiber and non-nutrient secondary metabolites that have been reported to be beneficial to human health (Aune et al. 2011). The protein content of fava beans is about twice that of cereal crops and several times that of root tubers (Wu Leung et al., 1968). However, the biological value of untreated fava beans is negatively affected by the presence of anti-nutritional factors such as the favism-inducing factors (aglycones: divicine and isouramil produced from metabolism of vicine and convicine; Jamalain et al., 1999; Frank, 2005; Luzzatto and Arese, 2018). The ingestion of untreated fava beans is associated with the hemolytic disease favism in certain glucose-6-phosphate dehydrogenase-deficient humans (Alkhaaldi et al., 2014; Eghbal et al., 2012; Kavehmansh et al., 2016; Skold et al., 2017; Frank, 2005; Luzzatto and Arese, 2018; Al-Dubai et al. 2021; Xin et al., 2022; Diegues et al., 2022). Favism is a condition causing red blood cells to break down in response to certain medications, infections, or other stressors. Typically, it is described as a condition when someone has a very severe reaction to fava beans and are at risk of acute hemolysis, in which their red blood cells break down faster than the body can replace them. Typically, glucose-6-phosphate dehydrogenase (G6PD) deficiency is an inherited condition usually occurring in males of African and Mediterranean descent.

Safety Data

Fava Beans and/or Fava Bean Protein

Given the long history of global human consumption of fava beans as food (and the protein contained therein), the safety of the fava bean protein ingredient derived from them is supported by data both on historical consumption and general lack of toxicity. As would be expected for a food that has been consumed by humans for centuries, fava beans and fava bean proteins have not been subjected to traditional toxicology studies. Given the publicly available information and data on the safety of fava beans and fava bean proteins, as well as the scientific efforts to remove anti-nutritional compounds from fava beans and fava bean protein isolates intended for animal and human consumption, the conduct of toxicity studies was considered unnecessary and not an ethical use of animals. A summary of publicly available safety-related information for fava beans and fava bean proteins is presented below.

Animal Studies

Macarulla et al. (2001) examined the effects of the whole fava bean seed versus a fava bean protein isolate on the cholesterol metabolism of hypercholesterolemic male Wistar rats. Rats (three groups; n=10) were fed *ad libitum* either high-fat diets rich in cholesterol-containing casein, whole seeds of *Vicia faba* (252 g protein/kg seed), or the protein isolate of fava beans (773 g protein/kg protein isolate) as a protein source for a period of 2 weeks. Analyses of serum, liver, and feces, as well as of the activity of hepatic 3-hydroxy-3-methylglutaryl (HMG)-CoA reductase, were assessed employing enzymatic methods. Rats fed *Vicia faba* diets had significantly lower body weights and energy intakes than did rats fed on casein diets. The whole-seed diet induced a significant reduction in plasma triacylglycerol. Feeding rats on diets containing fava bean seeds, or

the protein isolate, produced a significant decrease in plasma (LDL+VLDL)-cholesterol but not in HDL-cholesterol. Hepatic cholesterol and triacylglycerol were also reduced. The authors reported that the hypocholesterolemic effects of *Vicia faba* resulted from an increase in steroid fecal excretion. The authors concluded that the fava bean-protein isolate had a positive effect in improving the metabolic alterations induced by feeding with hypercholesterolemic diet compared to casein. No treatment-related adverse effects were noted.

Schmandke et al. (2000) conducted four-week feeding studies of several different processed dehulled fava bean materials in male and female Shoen:Wist/II rats. Treatment groups included (1) untreated beans, (2) beans steamed for 5 minutes at 120°C, (3) beans steamed for 15 min at 120°C, (4) beans soaked for 10 hours and heated for 2 minutes at 180°C by oil deep frying, and (5) a fava bean protein isolate. Thirty rats (15/sex) were used per group. Gradually increasing levels of the test materials were fed for 4 weeks, replacing the same amounts in the control diets. The fava bean protein isolate was fed at dose levels of 20, 40, and 80 g/kg. The composition of the protein isolate was approximately 81% protein, 6% fat, and 3% carbohydrate. The low dose of 20 g/kg was identified as the no-observed-effect level (NOEL) for female rats consuming the protein isolate (group 5). An increase in adrenal weights was noted at the low- and mid-dose levels for male rats and at the high dose of 80 g/kg for both sexes. In addition, an increase was noted in kidney weights and serum lymphocytes of female rats at the mid- and high-dose levels. No changes were noted in the weight of the spleen, liver or thymus. Heat treatment of the fava bean material (soaked and deep-fried dehulled faba beans) resulted in no toxicological effects at 50 g/kg doses (groups 3 and 4).

Human Clinical Trials

Nuritas has conducted clinical trials with the fava bean protein hydrolysate ingredient and no adverse effects were noted in either study.

1. In a randomized, controlled clinical trial (Weijzen et al., 2023), study participants were randomly allocated to ingest either 10 g of *Vicia faba* protein hydrolysate ($n=15$) or an isonitrogenous control (milk protein concentrate; MPC; $n=15$) twice daily throughout the study protocol (Trial # NTR645²). Creatinine, gamma-glutamyltransferase (GGT), aspartate aminotransferase (AST), alanine aminotransferase (ALT), lactate dehydrogenase (LDH), creatine kinase (CK), triglycerides (TG), high-sensitivity C-reactive protein (hs-CRP), free fatty acids (FFA), insulin and glucose concentrations were assessed in serum samples collected at baseline, prior to and following immobilization (i.e., one leg knee cast), and following remobilization. Data were analyzed for 20 participants ($n=9$ MPC, $n=11$ fava bean protein hydrolysate). Data for 10 participants were excluded due to missing samples or participants not being fasted. There were no changes over time and between groups for GGT, AST, ALT, LDH, CK, TG, hs-CRP, FFA, and insulin (all $P > 0.05$). Creatinine concentrations were lower following immobilization when compared to baseline and prior to immobilization

² www.clinicaltrials.gov

(both $P < 0.03$), but not different when compared to post-recovery. Creatinine was higher in the fava bean protein hydrolysate group when compared to the MPC group ($P < 0.001$). Glucose concentrations were lower following immobilization when compared to baseline ($P = 0.040$), with no differences between groups ($P > 0.05$). The investigators therefore concluded that a serving size of 20 g/day represented a safe level.

2. A randomized, placebo-controlled, double-blind trial of Elio™ (fava bean protein hydrolysate) administered to participants ($n=30$) at 2.4 grams/day assessed its effects on post-exercise strength recovery and increases in markers associated with muscle injury and exertion. (Trial # NCT05159375³). Following initial strength testing on day 0, subjects were administered either placebo or NPN_1 (2.4 g/day). On day 14, muscular fatigue and delayed onset muscle soreness (DOMS) was induced using resistance exercise. Strength recovery and fatigue were measured on day 16 and 17. Biomarker analysis was performed on blood samples collected prior to DOMS-inducing routine and 0, 2, 48 and 72-hours following completion of the routine. Fava bean protein hydrolysate supplementation significantly improved strength recovery post-resistance exercise compared to placebo over the 72-hour recovery period ($P=0.027$), measured by peak torque per body weight. Muscle fatigue was significantly reduced at both 48- and 72-hours post-resistance exercise ($P=0.004$), as was myostatin expression ($P=0.006$). A concomitant increase in other acute markers regulating muscle protein synthesis, regeneration and myoblast differentiation was also observed. The fava bean protein hydrolysate significantly improved strength recovery and restoration, reduced fatigue, and positively modulated alterations in markers related to muscle homeostasis (Kerr et al. 2023).

Anti-Nutritional Components

It has been long recognized that fava beans contain “anti-nutritional” compounds. Those that are of most concern are vicine (2,6-diamino-4,5-dihydropyrimidine 5-(beta-D-glucopyranoside) and convicine (2,4,5-trihydroxy-6-aminopyrimidine 5-beta-D-glucopyranoside), both of which are glycosidic aminopyrimidine derivatives. The vicine and convicine content of raw fava beans have been reported to range from 0.02% to 1.46% (dry weight basis) (Khamassi et al., 2013). Vicine and convicine are hydrolyzed in the fresh fava bean by β -glucosidase to form the aglycones divicine and isouramil (Figure 2, below). As reviewed in GRN 879, these aglycones are the compounds responsible for “favism,” by causing the oxidation of glutathione in red blood cells that cannot be sufficiently reversed in affected individuals who are mostly males. The erythrocytes become rigid, experience aggregation of their proteins, produce methemoglobin and have their enzymes inactivated due to oxidative damage that cannot be controlled or reversed. The damaged red blood cells are then removed by macrophages in the spleen and liver, which could lead to a potentially fatal hemolytic anemia.

³ [Effects of Protein Hydrolysate Supplementation Influencing Strength Recovery and Expression of Muscle Damage Markers Post-exercise - Full Text View - ClinicalTrials.gov](#)

For this reason, potential adverse reactions to fava bean ingestion are limited to a small, but significant, number of males with this X-linked genetic disease. There are many genetic variants of G6PD deficiency, with one common variant observed primarily in Africans and African-Americans and another in people with Mediterranean origins (e.g., Greeks, Italians, Armenians, several Semitic populations and others). Some Asian and middle eastern populations are also affected. G6PD deficiency in the U.S. is observed in approximately 10% of African-American males (Frank, 2005).

In the European, Middle Eastern, African, and Chinese areas where individuals prone to favism live, it is estimated that one meal can result in an intake of 2000 mg of vicine and 1000 mg of convicine (WHO, 1992). Numerous cases of favism related to consumption of raw, unprocessed fava beans by G6PD-deficient individuals have been reported (Alkhaaldi et al., 2014; Eghbal et al., 2012; Kavehmansh et al., 2016; Skold et al., 2017; Luzzatto and Arese, 2018; Al-Dubai et al. 2021; Xin et al., 2022).

The toxicity of vicine, convicine, divicine, and isouramil are discussed below, and the concentrations of the anti-nutrients vicine and convicine in three lots of the fava bean protein product are presented in Table 13. It should be noted that low-vicine fava bean seeds exist and can result in lower vicine/convicine levels such as those in Batch no. sh_FDE4MQ in Table 13 below. Because the seed type cannot be controlled by Nuritas, the potential exposure to the antinutritionals presented below, represents a worst-case calculation. The total amount of convicine and vicine in regular fava bean varieties is 6–14 mg/g (Arese et al., 2007).

Table 13. Alkaloid concentration in three lots of fava bean protein hydrolysate (mg/g)*

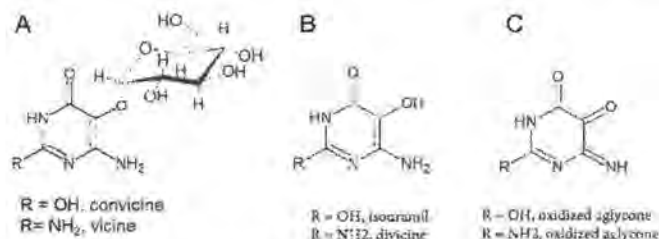
Alkaloid	Batch No. sh_CU6G3M	Batch No. sh_YO75MA	Batch No. sh_FDE4MQ
Vicine	14.1	11.3	0.54
Convicine	4.7	4.1	0.155
Sum	18.8	15.4	0.695

*by inverse phase UPLC method

Arese et al. (2007) stated that the levels of vicine and convicine in FEVITA (a low vicine/convicine protein ingredient) were 10 – 20x lower than traditional cultivars (i.e., 6–14 mg/g). A 10-fold reduction results in 0.6–1.4 mg/g and 0.3–0.7 mg/g for a 20-fold reduction. In the Arese et al. (2009) study, seven hemizygous males with G6PD deficiency used the FEVITA fava bean, which is low in vicine and convicine due to reduced beta-glucosidase in the FEVITA cultivar which prevents conversion of vicine and convicine to divicine and isouramil. The subjects were fed 450 g of the FEVITA fava bean which contained 135 - 315 mg of vicine and convicine. Given the worst-case vicine and convicine level of 18.8 mg/g in the proposed ingredient, times the mean and 90th percentile (users only) intakes of 6.4 and 12.4 g/day, respectively, results in an intake of approximately 120 mg and 233 mg vicine and convicine in the diet, amounts that are in the range of dose levels employed in the Arese et al. (2009) study in which no hemolytic

effects were reported. Furthermore, for children 6-18 years of age, employing the worst case vicine and convicine levels of 18.8 mg/g in the proposed ingredient and the per user mean and 90th percentile intakes of 5.6 and 11.2 g/day for 6-18 years of age, results in an intake of approximately 105 mg and 210 mg vicine and convicine respectively in the diet, similar to that of the intake values for the total population, 2+ years of age.

Figure 2. Chemical structures of (A) vicine and convicine, (B) their aglycones, divicine and isouramil, and (C) suggested oxidized aglycones (Pulkkinen et al., 2016)



Levodopa, also known as L-DOPA or L-3,4-dihydroxyphenylalanine, is a chemical precursor to dopamine and a common treatment for Parkinson's disease. L-DOPA naturally occurs in fava beans, but its levels decrease significantly in heated/cooked beans (Carador-Martinez et al., 2012; Multari et al, 2015). The level of L-DOPA in the fava bean protein hydrolysate was determined to be approximately 0.14% or 1400 mg/kg. If a person was to consume the 90th percentile (users only) intake of 12.4 g/day, the maximum calculated exposure is approximately 17 mg of levodopa, which is approximately 5.7% of the typical recommended daily dose (300 mg) and approximately 1.1% of the maximum recommended daily dose (1500 mg) for the treatment of Parkinson's disease. There is no prescribed dose of L-DOPA in children. For children 6-18 years of age, employing the L-DOPA level of 0.14% (1400 mg/kg) in the proposed ingredient and the per user 90th percentile intake for the 6-18 year old age group of 11.2 g/day, results in an intake of approximately 15.68 mg of L-DOPA. Levodopa has been studied in children with Angelman Syndrome (AS). In an open-label, dose-range finding study in children ages 4- 12 years with AS (Tan et al., 2018), various dosages (2 mg/kg bw/day, 5 mg/kg bw/day, 10 mg/kg bw/day, and 15 mg/kg bw/day) used (ClinicalTrials.gov Identifier: NCT00829439). The doses were chosen based on the parameters at which therapeutic effects (plasma levodopa concentration between 1 and 3 µg/ml) were observed in adults with Parkinson's. None of the participants who received less than 10 mg/kg bw/day of Levodopa achieved a plasma concentration above 1 µg/ml. Based on these results, a multi-center, double-blind, randomized, placebo-controlled one-year trial of Levodopa/Carbidopa with either 10 or 15 mg/kg bw/day of Levodopa in children ages 4-12 with AS was conducted. The number of reported adverse events, including the more serious ones, was similar in both groups, but none were related to treatment with Levodopa. The worst-case exposure to L-DOPA of 15.68 mg is a fraction of the low-dose of 2 mg/kg bw/day.

Animal Studies

Hussein (2012) reported acute oral LD50 values for vicine and divicine isolated from fava beans in adult male albino rats of 2100 and 1950 mg/kg bw, respectively. Oral administration of vicine to 10 rats/each dose/each group at doses of 700, 1400, 2100, 2800, 3500, and 4200 mg/kg bw resulted in 0, 2, 6, 8, 9, and 10 deaths, respectively. Oral administration of divicine in doses of 500, 1000, 1500, 2000, 2500, and 3000 mg/kg bw resulted in 0, 1, 2, 5, 8, and 10 deaths, respectively. Possible causes of death included heart failure that can occur by malfunction, acute hypoglycemia, and/or hepato-damage. Oral administration of vicine and divicine orally to rats at doses of 50 - 200 mg/kg bw for ten days did not produce significant changes in liver enzymes (SGOT, SGPT, ALP, LDH, γ -GT), serum TBARs, and blood enzyme (GSH, SOD, and GPx) levels. Histopathological results showed no change in liver morphology in rats treated with either vicine or divicine.

Koreim et al. (2008) investigated the effects of oral administration of convicine in Sprague-Dawley rats. The study design included a control group and two treatment groups (five male and five female rats/group, respectively). The two treatment groups were given oral doses of convicine (20 mg/kg bw) for 15 or 30 days. Convicine produced significant decreases ($p < 0.05$) in red blood cell counts and hemoglobin content. The serum albumin/globulin ratio was decreased significantly, and there were significant increases in serum and liver total protein, serum bilirubin, globulin, and iron levels. The study indicated that convicine produced alterations in blood similar to those observed in favism.

Human Studies

Arese et al. (2009) conducted a clinical trial to examine the safety of fava beans with low levels of the anti-nutritionals vicine and convicine (V/C) in seven hemizygous severely G6PD-deficient male subjects ($< 5\%$ normal enzyme activity). The subjects were fed 450 g/70 kg bw of fresh, raw, de-hulled beans, an amount known to exceed the normal quantity of fava bean consumption (i.e., 150 g/70 kg bw). This represents consumption of 135 - 630 mg of vicine plus convicine (i.e., 0.3 -1.4 mg vicine plus convicine/g times 450 g). Blood was drawn 24 hours before ingestion, and then 8-10, 24, and 48 hours after ingestion. Test parameters included 1) red blood cell (RBC) levels of reduced glutathione (GSH) (which is always decreased in favism); 2) markers indicating modification of RBC membranes (or oxidative stress), 3) formation of membrane adducts with 4-hydroxynonenal (4-HNE; a sensitive indicator of lipid peroxidation that accompanies oxidant insult), and 4) measurement of RBC filterability, a sensitive parameter of RBC deformability that is found during favism. In addition, RBC count, RBC volume, serum hemoglobin, hematocrit, and haptoglobin levels, as well as vitamin status, liver status, thyroid status, diabetes, and iron, were assessed to ensure that deficient subjects were without associated pathologies. Study results gave no indication of hemolysis as shown by constant levels of RBC counts, hematocrit level, hemoglobin parameters, percentage of reticulocytes, and level of bilirubin; no deficiencies in clinical chemistry results; and no change in the steady-state level of GSH. All other study parameters were stable over

time (e.g., 4-HNE adducts, autologous IgG, RBC deformability). The authors demonstrated that no parameters indicative of hemolysis or oxidative RBC damage were changed in the short-term (8–10 hours) or long-term (48 hours) after ingestion of approximately 10-fold greater than the normal amount of fava beans (low tannin, low-V/C fava beans). This study provided important clinical evidence in hemizygous severely G6PD-deficient male subjects that low-tannin, low-V/C fava beans are safe for consumption.

Allergy

Mur Gimeno et al. (2007) reported a case of an allergic reaction to bread that included fava bean (*Vicia faba*) flour. A 25-year-old woman presented with retrosternal and pharyngeal oppression and itching in the tongue and pharynx 5 minutes after eating the suspected bread. The subject had been allergic to legumes from the age of 8. A prick-by-prick test with the bread and a broad bean extract were positive. Additional immunoglobulin E (IgE)-related clinical testing supported the existence of allergens from broad beans in the bread that caused a type-I reaction in a legume-allergic patient.

Kumar et al. (2014) conducted studies to determine whether broad-bean proteins elicit allergic responses due to the presence of clinically relevant allergenic proteins. A simulated gastric fluid (SGF) assay and IgE immunoblotting were employed to identify pepsin-resistant and IgE-binding proteins. The allergenicity of broad beans was assessed in allergic patients, BALB/c mice, splenocytes, and RBL-2H3 cells. Eight broad-bean proteins of approximate molecular weight 70, 60, 48, 32, 23, 19, 15, and 10 kilodaltons that remained undigested in SGF, showed IgE-binding capacity. Of 127 allergic patients studied, broad-bean allergy was evident in 16 (12%). Mice sensitized with broad bean showed increased levels of histamine, total and specific IgE, and severe signs of systemic anaphylaxis compared with controls. Enhanced levels of histamine, prostaglandin D₂, cysteinyl leukotriene, and beta-hexosaminidase release were observed in the primed RBL-2H3 cells following broad-bean exposure. The authors concluded that broad-bean proteins have the ability to elicit allergic responses.

Damiani et al. (2011) reported a prolonged, and likely very high, occupational exposure to fava beans that resulted in hypersensitivity in a farmer. There are many known situations where very high occupational exposures to substances can induce hypersensitivity that is absent in typical consumer exposure to much lower levels with intermittent exposure. Consumer exposure via food is very unlikely to replicate this farmer's situation.

Jensen et al. (2008) performed allergy cross-reactivity testing on ten children with moderate-to-severe peanut allergy. Broad beans produced positive Skin Prick Test results in only two of the 10 children. One had a 3.5 mm wheal that barely exceeded the minimal positive level (3mm). The other child showed a significant positive result with broad bean and every other food protein tested indicating a high propensity to show a cross-reactivity response. The result simply confirmed that cross-reactivity in peanut allergic individuals may rarely occur with fava bean protein just as it might with other legumes. Although

fava beans did show cross-reactivity in some peanut-allergy patients, additional tests showed that fava bean proteins were about 1000-fold less potent than peanut proteins in inducing histamine release (Figure 1, Jensen et al., 2008).

A few reported cases of hemolytic anemia have been reported in breastfed infants of women consuming raw or cooked whole fava beans containing high levels of vicine and convicine. The concentrations of these same alkaloids in the proposed fava bean protein hydrolysate product that is the subject of this GRAS dossier are very low. Therefore, it is unlikely that significant levels of vicine and convicine would be present in breast milk following consumption of the specified foods containing the proposed protein hydrolysate ingredient that would represent a hazard to either the mother or infant.

The potential for fava bean proteins to cause an immune response is rare but consistent with similar, known allergic responses to other legumes. Although fava beans are not listed as one of eight major allergen groups by the FDA under the Food Allergen Labeling and Consumer Protection Act of 2004 (Public Law 108-282, Title II), the fact that the allergenicity of fava bean protein has been shown clinically (Mur Gimeno et al., 2007; Kumar et al., 2014) clearly indicates that labeling of the presence of fava bean protein is warranted and recommended. More importantly, given the incidence of G6PD deficiency in the US, consumers may not expect fava bean protein to be present in their food. The GRAS Panel recommends that the ingredient labeling for the fava bean protein hydrolysate product clearly state that it contains “fava bean protein” and individuals who wish to avoid fava bean or fava bean protein consumption for any reason would be able to identify the presence of a fava bean–derived ingredient.

Other Protein-Related Safety Concerns

Protein Intake and Toxicity

IOM recommends that adults not consume more than twice the 0.8 g/kg protein RDA per day. However, physically active persons on normal diets can easily exceed this level, and individuals involved in bodybuilding ingest much higher levels of protein (WHO, 2002). WHO (2002) recommends body-weight-based protein consumption rates for both genders. For example, the safe protein consumption level for a 40-kg adult is 33 g/day, and that for an 80-kg adult is 66 g/day.

Kidney Function

Dietary protein is known to alter renal function. Increased protein intakes lead to increased excretion of urea and creatinine, due to increased renal blood flow causing a higher glomerular filtration rate. Excess protein intake has been found to advance chronic kidney disease due to increased glomerular pressure and hyperfiltration (Martin et al., 2005; WHO, 2002). Martin et al. (2005) state that, although protein restriction may be indicated in treatment of existing kidney disease, the existing evidence does not indicate an adverse effect of high protein consumption on renal function in healthy individuals who for centuries have consumed high-protein Western diets. In addition, several studies

indicate that hyperfiltration, the reported mechanism for kidney effects, is a normal adaptive response to increased demands for renal clearance due to higher nitrogen load.

Martin et al. (2005) defined a “high protein diet” as a daily intake of ≥ 1.5 g/kg-day, nearly double the current recommended intake level set by IOM. In persons with decreased renal function, a high-protein diet that leads to a renal solute load in excess of the kidneys’ excretory functions can contribute to progressive kidney failure. The intake of the fava bean protein ingredient from the intended uses in select foods proposed in this dossier will result in daily intake levels below those associated with effects on renal function. As to the issue of kidney stone formation, it is recommended that those who are at risk should consume the recommended safe level of protein (0.83 g/kg-day), preferably from vegetable sources, but not high levels (>1.4 g/kg/day) (WHO, 2002).

Calcium Balance/Bone Health

Excess protein intake can adversely affect the body’s calcium balance and calcium concentration in bone. Persons consuming high-protein diets are known to excrete increased amounts of urinary calcium, and doubling protein intake can increase urinary calcium excretion by 50%. WHO (2002) states that the existing scientific evidence indicates that dietary protein, when consumed as part of a well-balanced diet, is likely beneficial for bone, potentially even at dietary levels exceeding the recommended consumption rates.

Summary

The proposed uses and consumption of fava bean protein hydrolysate in the foods specified in Part 3 do not raise concerns regarding the noted protein safety-related outcomes.

Basis for the GRAS Determination

Introduction

The regulatory framework for determining whether a substance can be considered GRAS in accordance with section 201(s) (21 U.S.C. § 321(s)) of the Federal Food, Drug, and Cosmetic Act (21 U.S.C. § 301 et. Seq.) (“the Act”) is set forth at 21 CFR 170.30, which states:

General recognition of safety may be based only on the view of experts qualified by scientific training and experience to evaluate the safety of substances directly or indirectly added to food. The basis of such views may be either (1) scientific procedures or (2) in the case of a substance used in food prior to January 1, 1958, through experience based on common use in food. General recognition of safety requires common knowledge about the substance throughout the scientific community knowledgeable about the safety of substances directly or indirectly added to food.

General recognition of safety based upon scientific procedures shall require the same quantity and quality of scientific evidence as is required to obtain approval of a food additive regulation for the ingredient. General recognition of safety through scientific procedures shall ordinarily be based upon published studies, which may be corroborated by unpublished studies and other data and information.

These criteria are applied in the analysis below to determine whether the use of the fava bean protein hydrolysate ingredient in specified human foods identified in this dossier and that is the subject of this GRAS determination is GRAS based on scientific procedures. All data relied upon in this GRAS determination are publicly available and generally known, and therefore meet the “general recognition” standard under the Federal Food, Drug, and Cosmetic Act (FFDCA).

Safety Determination

The subject of this GRAS determination is the use of fava bean protein hydrolysate as an alternative source of dietary protein for addition to processed foods. Humans have consumed fava beans and other legumes, as well as proteins from these sources, for centuries, along with proteins from many food sources such as meats, dairy, fruits, vegetables, nuts, and seeds. Other natural, plant-based sources of protein concentrates that have been safely consumed for years include soy, canola, potato, wheat, whey, and mung bean.

IOM (2005) recommends that adults consume a minimum of 0.8 g protein/kg and has set a range for acceptable protein intake of 10%–35% of daily calories. In the US, the recommended daily allowance of protein is 56 g/day and 46 g/day for adult men and women (>19 years of age), respectively. However, it should be noted that the RDA does not represent an upper limit of consumption.

As described in numerous GRAS Notifications, including GRN No. 386 for canola protein isolate and hydrolyzed canola protein isolate, the typical uses of protein for enrichment of foods include bakery products, snack foods, ready-to-drink beverages, soups and nutritional beverages, high-protein drinks and milkshakes, powdered nutritional/protein beverages, nutrition bars, vegetarian food products, meat analogues, processed meat products, dairy and imitation dairy products, and meal replacements/nutrition bars.

Nuritas’ proposed fava bean protein hydrolysate is intended only to be an alternative source of protein for use in certain specified foods identified in this dossier. As was concluded in numerous other GRAS notifications for plant-based dietary proteins for which FDA issued “no objection” letters, it is not realistic to expect that the actual consumption of foods containing fava bean protein hydrolysate would result in daily protein consumption greater than the existing RDA for protein. It is reasonable to expect that most of the population’s daily intake of protein will remain in the form of unprocessed foods, including meat, poultry, fish, and legumes (FDA, 2010, 2011).

Fava beans are a rich source of fiber and non-nutrient secondary metabolites shown to be beneficial to human health. Their protein content is about twice that of other cereals and several times that of root tubers. However, the biological value of untreated fava beans is negatively affected by the presence of anti-nutritional factors such as the favism-inducing factors (the aglycones: divicine and isouramil, from vicine and convicine), and the ingestion of untreated fava beans is associated with the hemolytic disease “favism” in certain glucose-6-phosphate dehydrogenase-deficient humans.

In the European, Middle Eastern, and Chinese areas where individuals prone to favism live, and where fava beans, and especially the immature pods, are consumed, it is estimated that there is an intake of 2000 mg of vicine and 1000 mg of convicine in one meal (WHO, 1992). Numerous cases of favism related to consumption of raw, unprocessed fava beans by G6PD-deficient individuals have been reported (Alkhaaldi et al., 2014; Eghbal et al., 2012; Kavehmansh et al., 2016; Skold et al., 2017).

The total amount of convicine and vicine in a regular fava bean variety is 6–14 mg/g. Arese et al. (2007) stated that the levels of vicine and convicine in FEVITA (a low vicine/convicine protein ingredient) were 10 – 20x lower than traditional cultivars (i.e., 6–14 mg/g). A subsequent study by Arese et al. (2009) conducted in seven hemizygous males with G6PD deficiency used the FEVITA fava bean, which is low in vicine and convicine due to reduced beta-glucosidase in the FEVITA cultivar which prevents conversion of vicine and convicine to divicine and isouramil. The subjects were fed 450 g of the FEVITA fava bean which contained 135 - 315 mg of vicine and convicine. Given the vicine and convicine level of 18.8 mg/g in the proposed ingredient, and the mean and 90th percentile (users only) intakes of 6.4 and 12.4 g/day of fava bean protein hydrolysate, respectively, results in daily intakes of approximately 120 mg and 233 mg vicine and convicine in the diet (mean and 90th percentile), amounts that are in the range of dose levels employed in the Arese et al. (2009) study in which no hemolytic effects were reported. In addition, the maximum serving size of this proposed fava bean protein hydrolysate limits the amount of vicine and convicine in the final consumer food product.

The potential for fava bean proteins to cause an immune response is rare but consistent with similar, known allergens found in other legumes. Although fava beans are not listed as one of eight major allergen groups by the FDA under the Food Allergen Labeling and Consumer Protection Act of 2004 (Public Law 108-282, Title II), the fact that the allergenicity of fava bean protein has been shown clinically (Mur Gimeno et al., 2007; Kumar et al., 2014) suggests that labeling of the presence of fava bean protein is both warranted and recommended. More importantly, given the incidence in of G6PD deficiency in the US, consumers may not expect fava bean protein to be present in their food. Therefore, the GRAS Panel recommends that the ingredient labeling for the fava bean protein product clearly state that it contains “fava bean protein” and individuals who wish to avoid fava bean or fava bean protein consumption for any reason would be able to identify the presence of a fava bean–derived ingredient.

General Recognition of the Safety of Fava Bean Protein Ingredient

The intended use of the fava bean protein hydrolysate ingredient in human food has been determined to be safe through scientific procedures set forth in 21 CFR §170.3(b), thus satisfying the so-called “technical” element of the GRAS determination, based on the following:

- Fava bean protein hydrolysate is manufactured from commercially available fava beans, following current cGMP for food (21 CFR § Part 110). The raw materials and processing aids used in the manufacturing process are food grade and/or approved for use in food. The fava bean protein hydrolysate product has been characterized appropriately, contains approximately 60% protein, and meets appropriate food-grade specifications.
- Fava beans have been consumed as food (and the protein contained therein) for centuries, along with many other food sources of protein (e.g., meats, dairy, fruits, vegetables, nuts).
- The biological value of untreated fava beans is negatively affected by the presence of anti-nutritional factors such as the favism-inducing factors (the aglycones: divicine and isouramil produced from metabolism of vicine and convicine). The ingestion of untreated fava beans is associated with precipitation of the hemolytic disease “favism” in certain glucose-6-phosphate dehydrogenase-deficient humans.
- These anti-nutrients represent a known safety concern. Calculated daily intakes of the fava bean protein hydrolysate of approximately 120 mg and 233 mg vicine and convicine in the diet (mean and 90th percentile), represents amounts of vicine and convicine that are in the range of dose levels employed in the Arese et al. (2009) study in which no hemolytic effects were reported.
- The proposed uses of the fava bean protein hydrolysate ingredient in specified foods will provide an alternative to other dietary sources of protein and will not increase the overall consumption of protein but will simply provide another source of well-characterized protein for use in food products.
- Fava beans are a rich source of fiber and non-nutrient secondary metabolites shown to be beneficial to human health. The protein content is about twice that of other cereals and several times that of root tubers.
- FDA has reviewed published information and data on many protein products as part of GRAS notifications for animal and plant-based protein isolates and concentrates and subsequently issued “no objection letters.” Examples include GRN No. 26 (isolated wheat protein); GRN No. 37 (whey protein isolate and dairy product solids); GRN No. 168 (poultry protein); GRN No. 182 (hydrolyzed wheat gluten isolate; pea protein isolate); GRN No. 313 (beef protein); GRN No.

314 (pork protein); GRN 386 (canola protein isolate and hydrolyzed canola protein isolate); GRN No. 447 (potato protein isolates); GRN No. 575 (oat protein); GRNs No. 58, 608, and 788 (pea protein); GRN 879 (fava bean protein); and GRN 944 (rice protein hydrolysate).

- Given the long history of global human consumption of fava beans as food (and the protein contained therein), the safety of the fava bean protein hydrolysate ingredient derived from them is supported by a history of consumption and general lack of toxicity. As would be expected for a food that has been consumed by humans for centuries, fava beans and fava bean proteins have not been subjected to traditional toxicology studies. However, the available preclinical and clinical study data support its safe use as proposed.
- Concerns related to the allergenicity of fava bean protein as well as the effect of fava beans consumption by G6PD deficient individuals will be addressed through appropriate labeling of food products as containing fava bean protein and individuals who wish to avoid fava bean protein consumption would be able to identify the presence of a fava bean-derived ingredient.
- The body of publicly available scientific literature on the consumption and safety of fava beans and fava bean protein is sufficient to support the safety and GRAS status of the proposed fava bean protein hydrolysate ingredient.

Because this safety evaluation was based on generally available and widely accepted data and information, it also satisfies the so-called “common knowledge” element of a GRAS determination.

Determination of the safety and GRAS status of the fava bean protein hydrolysate ingredient for its intended uses in specified foods that is the subject of this self-determination has been made through the deliberations of a GRAS Panel of qualified experts convened by Nuritas and composed of Paul Damian, Ph.D., M.P.H., DABT, ERT, Carol A. Knight, Ph.D., and Stanley M. Tarka, Jr., Ph.D., F.A.T.S. These individuals are qualified by scientific training and experience to evaluate the safety of substances intended to be added to food. They have critically reviewed and evaluated the publicly available information summarized in this document and have individually and collectively concluded that the fava bean protein hydrolysate ingredient, produced in a manner consistent with cGMP and meeting the specifications described herein, is safe under its intended conditions of use. The Panel further unanimously concluded that the use of the fava bean protein hydrolysate ingredient in certain specified human foods is GRAS based on scientific procedures, and that other experts qualified to assess the safety of food and food ingredients for human consumption would concur with these conclusions. The Panel’s GRAS opinion is included as Exhibit 1 to this document.

It is also Nuritas’ opinion that other qualified scientists reviewing the same publicly available toxicological and safety information would reach the same conclusion. Nuritas has concluded that the fava bean protein hydrolysate ingredient is GRAS under the

intended conditions of use on the basis of scientific procedures; and therefore, it is excluded from the definition of a food additive and may be marketed and sold for its intended purpose in the U.S. without the promulgation of a food additive regulation under Title 21 of the CFR.

Nuritas is not aware of any information that would be inconsistent with a finding that the use of the fava bean protein hydrolysate ingredient in food for human consumption, meeting appropriate specifications, and used according to GMP, is GRAS. Recent reviews of the scientific literature revealed no potential adverse health concerns.

§170.250 Part 7, Supporting Data and Information

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APPENDIX A

COAs and Other Analytical Data

Certificate of analysis

Product name	PeptiStrong™
Product description	Fava bean (<i>Vicia faba</i>) plant (bean) based protein hydrolysate powder
Lot number	sh_CU6G3M
Best before	June 2022

<u>Test</u>	<u>Specification</u>	<u>Lot results</u>
pH	pH 5 – 7	pH 6.08
Moisture	≤ 10 %	5.26 %
Protein	55 – 65 %	58.98 %
Aerobic plate count	≤ 10 ⁶ CFU/g	300 CFU/g
Mould and yeast	≤ 1000 CFU/g	< 10 CFU/g
<i>E. coli</i>	≤ 20 CFU/g	< 10 CFU/g
<i>Salmonella</i> spp.	Not detected (CFU/25g)	Not detected
Coliforms	< 10 CFU/g	< 10 CFU/g
Mercury	≤ 0.05 ppm	<0.01 ppm
Arsenic	≤ 0.05 ppm	<0.01 ppm
Cadmium	≤ 0.05 ppm	0.015 ppm
Lead	≤ 0.05 ppm	<0.01 ppm
QC release date	N/A	05 August 2020

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Certificate of analysis

Product name	PeptiStrong™
Product description	Fava bean (<i>Vicia faba</i>) plant (bean) based protein hydrolysate powder
Lot number	sh_YO75MA
Best before	Dec 2023

<u>Test</u>	<u>Specification</u>	<u>Lot results</u>
pH	pH 5 – 7	pH 6.23
Moisture	≤ 10 %	6.31 %
Protein	55 – 65 %	56.45 %
Gluten	≤ 20 ppm	< 5 ppm
Aerobic plate count	≤ 10 ⁶ CFU/g	2,300 CFU/g
Mould and yeast	≤ 1000 CFU/g	< 100 CFU/g
<i>E. coli</i>	≤ 20 CFU/g	< 10 CFU/g
<i>Salmonella</i> spp.	Not detected (CFU/25g)	Not detected
Coliform	≤ 10 CFU/g	≤ 10 CFU/g
Mercury	≤ 0.05 ppm	<0.01 ppm
Arsenic	≤ 0.05 ppm	<0.01 ppm
Cadmium	≤ 0.05 ppm	0.011 ppm
Lead	≤ 0.05 ppm	<0.01 ppm
QC release date	N/A	15 July 2021


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Certificate of analysis

Product name	PeptiStrong™
Product description	Fava bean (<i>Vicia faba</i>) hydrolysate peptide complex
Batch number	sh_FDE4MQ
Best before	27 JUN 2025

<u>Test</u>	<u>Specification</u>	<u>Lot results</u>
pH	pH 5 – 7	pH 6.27
Moisture	≤ 10 %	4.87 %
Protein	55 – 65 %	60 %
Gluten	≤ 20 ppm	< 5 ppm
Aerobic plate count	≤ 10 ⁶ CFU/g	240 CFU/g
Mould and yeast	≤ 1000 CFU/g	< 10 CFU/g
<i>E. coli</i>	≤ 20 CFU/g	< 10 CFU/g
<i>Salmonella</i> spp.	Not detected (CFU/25g)	Not detected (25g)
<i>Coliform</i>	≤ 10 CFU/g	≤ 10 CFU/g
Mercury	≤ 0.05 ppm	< 0.013 mg/kg
Arsenic	≤ 0.05 ppm	0.014 mg/kg
Cadmium	≤ 0.05 ppm	0.015 mg/kg
Lead	≤ 0.05 ppm	< 0.007 mg/kg
QC release date	N/A	07 JUL 2023


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Certificate of analysis

Product name	PeptiStrong™
Product description	Fava bean (<i>Vicia faba</i>) plant (bean) based protein hydrolysate powder
Lot number	sh_CU6G3M
Best before	June 2022

Test

Vicine
Convicine

Test method

inverse phase UPLC
Inverse phase UPLC

Lot results

14.1 mg/g
4.7 mg/g



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Certificate of analysis

Product name	PeptiStrong™
Product description	Fava bean (<i>Vicia faba</i>) plant (bean) based protein hydrolysate powder
Lot number	sh_YO75MA
Best before	Dec 2023

Test

Vicine
Convicine

Test method

Inverse phase UPLC
Inverse phase UPLC

Lot results

11.3 mg/g
4.1 mg/g



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Certificate of analysis

Product name	PeptiStrong™
Product description	Fava bean (<i>Vicia faba</i>) hydrolysate peptide complex
Batch number	sh_FDE4MQ
Best before	27 JUN 2025

Test

Vicine
Convicine

Test method

Inverse phase UPLC
Inverse phase UPLC

Lot results

0.54 mg/g
0.155 mg/g



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Mycotoxin Screen	Results			Units	MRL	Recovery %	Method	
	Batch:	sh_3X7JA0	sh_CU6G3M					sh_YO75MA
Aflatoxin B1(recovery corrected)		<0.5	<0.5	<0.5	µg / kg	-	82	AM/R/999
Aflatoxin B2(recovery corrected)		<0.5	<0.5	<0.5	µg / kg	-	84	AM/R/999
Aflatoxin G1(recovery corrected)		<0.5	<0.5	<0.5	µg / kg	-	83	AM/R/999
Aflatoxin G2(recovery corrected)		<0.5	<0.5	<0.5	µg / kg	-	89	AM/R/999
Total Aflatoxin(recovery corrected)		<2	<2	<2	µg / kg	-	-	AM/R/999

Amino Acid Profiles	Abbrev	Code	Results			Units
			sh_3X7JA0	sh_CU6G3M	sh_YO75MA	
Alanine	Ala	A	2.15	2.26	2.14	g / 100g
Arginine	Arg	R	5.44	5.14	5.11	g / 100g
Aspartic acid	Asp	D	6.00	6.27	5.96	g / 100g
Cysteine	Cys	C	0.60	0.53	0.53	g / 100g
Glutamic acid	Glu	E	9.01	9.36	8.80	g / 100g
Glycine	Gly	G	2.20	2.34	2.20	g / 100g
Histidine	His	H	1.43	1.46	1.34	g / 100g
Isoleucine	Ile	I	2.38	1.69	1.57	g / 100g
Leucine	Leu	L	3.90	3.85	3.59	g / 100g
Lysine	Lys	K	3.66	3.63	3.49	g / 100g
Methionine	Met	M	0.41	0.41	0.40	g / 100g
Phenylalanine	Phe	F	2.35	2.26	2.09	g / 100g
Proline	Pro	P	2.51	2.55	2.37	g / 100g
Serine	Ser	S	2.99	3.24	2.98	g / 100g
Threonine	Thr	T	2.09	2.07	1.89	g / 100g
Tyrosine	Tyr	Y	1.79	1.91	1.77	g / 100g
Valine	Val	V	2.54	1.93	1.77	g / 100g

GCMS/MS Pesticide Screen

	Results			Units	MRL	Rec %	Method
	Batch: sh_3X7JA0	sh_CU6G3M	sh_YO75MA				
1,4-dimethylnaphthalene	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Acephate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Acetochlor	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Acibenzolar-S-methyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Aclonifen	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Acrinathrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Alachlor	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Allethrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Ametryn	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Anthraquinone	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Atraton	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Atrazine	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Azaconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Azinphos ethy	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Azinphos methyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Azoxystrobin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Benalaxyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Benfluralin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Bentazone (parent only)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Bifenox	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Bifenthrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Biphenyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Bitertanol	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Boscalid	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Bromocyclen	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Bromophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Bromophos-ethyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Bromopropylate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003

Bromuconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Bupirimate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Buprofezin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Butafenacil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Butralin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Cadusaphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Carbophenothion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlorfenvinphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chinomethionate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
cis Chlordane	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
trans Chlordane	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlordane (sum of cis & trans isomers)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlordecone	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlorfenapyr	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlorfenson	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlormephos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlorobenzilate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chloroneb	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chloropropylate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlorpropham	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlorpyrifos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlorpyrifos-methyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlorthal-dimethyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlorthion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Chlozolate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Climbazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Clodinafop-propargyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Cloquintocet-mexyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Crimidine	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Cyanazine	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Cyanofenphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003

Cyanophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Cyfluthrin (sum of isomers)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Lambda Cyhalothrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Cypermethrin (sum of isomers)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Cyproconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Cyprodinil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
o,p'-DDD	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
p,p'-DDD	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
o,p'-DDE	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
p,p'-DDE	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
o,p'-DDT	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
p,p'-DDT	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
DDT (sum of pp'-DDE,pp'-DDD,op'-DDT & pp'-DDT)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Deltamethrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Desmetryn	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Diallate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Diazinon	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dichlobenil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dichlofenthion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dichlorvos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Diclobutrazol	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Diclofop-methyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dicloran	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dicofol	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dicrotophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Aldrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dieldrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Sum of Aldrin and Dieldrin (expressed as Dieldrin)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Difenoconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dimethomorph	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dioxabenzofos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003

Diphenylamine	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Dipropetryn	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Ditalimfos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Edifenphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Endosulfan - Alpha	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Endosulfan - Beta	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Endosulfan sulphate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Endosulfan Sum Total)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Endrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
EPN	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
EPTC	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Etaconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Ethion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Ethofumesate (parent only)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Ethoprophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Etiozazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Etridiazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Etrimfos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Flamprop isopropyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Famphur	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenarimol	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenchlorphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenchlorphos oxon	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenchlorphos (sum of Fenchlorphos & Fenchlophos oxon)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenitrothion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenpiclonil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenpropathrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenpropimorph	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenson	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fensulfothion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fenthion (parent only)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003

Fenvalerate (sum of isomers)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fipronil (parent only)	<0.005	<0.005	<0.005	mg / kg	-	-	AM/R/1003
Fluazifop-p-butyl (parent only)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fluchloralin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Flucythrinate (sum of isomers)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fludioxonil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Flumetralin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Flumioxazine	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fluopyram	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Flurprimidole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Flusilazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fluthiacet-methyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Flutolanil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Flutriafol	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
tau-Fluvalinate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fluxapyroxad	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Fonofos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Furalaxyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Furathiocarb	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Hexachlorobenzene	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Hexachlorocyclohexane-alpha	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Hexachlorocyclohexane-beta	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Hexachlorocyclohexane-delta	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Hexachlorocyclohexane-gamma	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Heptachlor	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
trans Heptachlor epoxide	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
cis Heptachlor epoxide	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Heptachlor (sum of heptachlor & heptachlor epoxide)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Heptenophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Hexaconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Imazalil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003

Iodofenphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Iprodione	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Isazophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Isobenzan	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Isodrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Isofenphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Isoprocab	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Isoxadifen-ethyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Isoxathion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Kresoxim-methyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Leptophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Malaoxon	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Malathion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Sum of malathion and malaoxon (expressed as malathion)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
MCPA-thioethyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Mecarbam	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Mefenapir-diethyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Mepaniprim	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Metalaxyl and metalaxyl-M (sum of isomers)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Metazachlor (Parent Only)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Metconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Methacrifos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Methamidophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Methidathion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Methoxychlor	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Metrafenone	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Metribuzin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Mevinphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Mirex	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Monocrotophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Myclobutanil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003

Napropamide	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Nitralin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Nitrofen	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Nitrothal-isopropyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Nuarimol	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Ofurace	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Omethoate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Oxadixyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Oxyfluorfen	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Paclobutrazol	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Parathion-ethyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Parathion-methyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pebulate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Penconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pendimethalin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pentachloroanisole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pentachlorobenzene	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pentachlorophenol	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Permethrin (sum of isomers)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Perthan	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Phenothrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Phenthoate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Phosalone	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Phosfolan	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Phosmet (parent only)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Phosphamidon	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Piperonyl Butoxide	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pirimicarb	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pirimiphos - ethyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pirimiphos-methyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Procymidone	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003

Profenofos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Profluralin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Prometryn	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Propachlor	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Propargite	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Propazine	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Propetamphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Propham	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Propiconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Propyzamide	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Prothiofos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pyrazophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pyridaphenthion	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pyrimethanil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Quinalphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Pentachloroaniline	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Quintozene	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Quintozene (sum of quintozene & pentachloroaniline)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Resmethrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Silafluofen	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Simazine	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Sulfotep	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Sulprofos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Tetrachlorvinphos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Tebuconazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Tecnazene	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Tefluthrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Terbacil	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Terbufos (sum of terbufos, terbufos sulfone & terbufos sulfoxide)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Terbumeton	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003

Terbutylazine	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Terbutryn	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Tetradifon	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Tetramethrin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Tetrasul	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Thiabendazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Thiometon	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Tolclofos-methyl	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Transfluthin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Triadimefon	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Triadimenol	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Tri-allate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Triazamate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Triazophos	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Trichloronate	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Trietazine	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Trifloxystrobin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Trifluralin	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Triticonazole	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Vinclozolin (parent compound only)	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003
Zoxamide	<0.01	<0.01	<0.01	mg / kg	-	-	AM/R/1003

APPENDIX B

Intake Assessment Report

Estimated Daily Intake of Fava Bean Protein Hydrolysate

NOVEMBER 10, 2022

Tox Strategies

Innovative solutions
Sound science

Estimated Daily Intake of Fava Bean Protein Hydrolysate

NOVEMBER 10, 2022

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Acronyms

CDC	Centers for Disease Control and Prevention
EDI	estimated daily intake
EPA	U.S. Environmental Protection Agency
g/kg-bw/day	grams per kilogram body weight per day
NHANES	National Health and Nutrition Examination Survey
RTD	ready to drink
USDA	U.S. Department of Agriculture
WWEIA	What We Eat in America

Executive Summary

ToxStrategies has conducted an intake assessment to estimate the mean and 90th percentile daily intake of the ingredient fava bean protein hydrolysate based on its intended use in foods. This assessment included selected foods that include dry-blend powders, ready-to-drink (RTD) beverages, and meat and dairy substitutes in which fava bean protein hydrolysate is intended to be used. After calculating serving sizes of selected foods using the What We Eat in America — Food Commodity Intake Database (WWEIA-FCID, 2005–2010) and analyzing dietary survey data from the National Health and Nutrition Examination Survey (NHANES), the total estimated daily intake (EDI) of fava bean protein hydrolysate and fava bean protein were calculated. The per-user mean and 90th percentile EDIs of the fava bean protein hydrolysate for the U.S. population ages 2 years and older were determined to be 6.44 and 12.4 g/day (0.0990 and 0.199 g/kg-bw/day), respectively. The per-capita mean and 90th percentile EDI of fava bean protein hydrolysate for the U.S. population ages 2 years and older were determined to be 0.92 and 2.88 g/day (0.014 and 0.042 g/kg-bw/day), respectively. Given these intake estimates, the fava bean protein mean and 90th percentile contributions for users 2 years and older are 3.86 g/day and 7.45 g/day (0.594 and 0.120 g/kg-bw/day). Per capita, the fava bean protein-only EDI mean and 90th percentile were 0.553 and 1.73, respectively, and 0.008 and 0.025 g/kg-bw/day.

1 Data

To calculate the fava bean protein hydrolysate EDI, information about its proposed uses in specified foods was combined with up-to-date, publicly available dietary intake survey data. Data sources are described in the following sections.

1.1 Proposed Uses and Use Levels of Fava Beans

Nuritas proposes to use fava bean protein hydrolysate in various food categories according to Table 1.

Table 1. Proposed uses and serving size contribution of fava bean protein hydrolysate

Food Category	Food Subcategory	Maximum Proposed Use Level (g/serving)
Dry-blend powders	Protein Powders	2.4 g
	Instant Breakfast	2.4 g
	Sports Drink Blends	2.4 g
Meal replacements	Drinks	2.4 g
	Bars	2.4 g

Food Category	Food Subcategory	Maximum Proposed Use Level (g/serving)
Dairy alternatives	Milk Analogs	2.4 g
	Grits	2.4 g
	Oatmeal	2.4 g
RTD beverages	Protein Shakes	2.4 g
	Sports Drinks	2.4 g
Nutritionally fortified confectionary	Hard Candies	2.4 g
	Gummies	2.4 g
Nutritionally fortified snack foods	Protein Bars	2.4 g
	Energy Bars	2.4 g
Baked goods	Cookies	2.4 g
	Biscuits	2.4 g
	Cereal Bars	2.4 g
Meat alternatives	Veggie Burger	2.4 g
	Meatless Fish Stick	2.4 g
	Meatless Pork/Beef	2.4 g

1.2 Dietary Survey Data

Dietary survey data were obtained from What We Eat in America (WWEIA), the dietary interview portion of the National Health and Nutrition Examination Survey (NHANES). NHANES is carried out in 2-year cycles (biennials) by the Centers for Disease Control and Prevention (CDC), to characterize the general health and nutritional status of children and adults across the U.S. The dietary intake survey portion of NHANES is a joint effort between CDC and the U.S. Department of Agriculture (USDA). All NHANES biennials for which dietary intake data were available were included in this analysis (2003–2004, 2005–2006, 2007–2008, 2009–2010, 2011–2012, 2013–2014, and 2015–2016).

The first day of the WWEIA dietary questionnaire was administered in person, in conjunction with the participants' interviews and examinations for the other NHANES lifestyle and laboratory assessments. The second day of the survey was collected via a phone interview at some point 3–10 days after the first survey day. Data collected during the dietary interview include foods consumed by the participant, encoded by a food code, and amounts eaten. Respondents who provided complete records for both days were designated as reliable by WWEIA; only reliable respondents who also provided body-weight data were considered in this analysis.

2 Methods

To estimate the intake of fava bean protein hydrolysate from its proposed uses in specified foods, ToxStrategies performed the following steps:

- Step 1: Identified foods and their components to which fava bean protein hydrolysate may be added
- Step 2: Calculated individual intake of fava bean protein hydrolysate for individual survey participants
- Step 3: Calculated population statistics estimating fava bean protein hydrolysate intake
- Step 4: Estimate protein contribution of fava bean protein hydrolysate addition.

Details of each step are provided in the following sections.

2.1 Identification of Foods and Their Components to which Fava Bean Protein Hydrolysate May Be Applied

To identify foods that are proposed to contain fava bean protein hydrolysate, ToxStrategies performed a thorough search of food codes reported in WWEIA. Food code descriptions from WWEIA were examined manually and flagged if they could be considered members of one or more of the food categories specified in Table 1. Food codes retained for further analysis are listed in the appendix.

2.2 Calculation of Individual Intake of Fava Bean Protein Hydrolysate for Individual Survey Participants

Only those respondents designated as reliable were included in this assessment.

Both days of the NHANES WWEIA dietary interviews from each biennial were analyzed. Participants' consumption of the fava beans was averaged over the two response days—i.e., $(\text{Day1 consumption} + \text{Day2 consumption})/2$. Consumption of fava bean protein hydrolysate was calculated using the grams of the relevant food consumed, as reported in NHANES, multiplied by the maximum proposed use level (2.4 g/serving) of fava bean protein hydrolysate in that food. For example, for the food “53720500, Snickers Marathon Protein Bar,” the total food weight is 80 g per serving. Thus, a survey participant who consumed this food would consume a proportion of 0.03 g of fava bean protein hydrolysate per gram of food.

2.3 Calculation of Population Statistics Describing Fava Bean Protein Hydrolysate Estimated Daily Intake

All NHANES biennials for which dietary intake data were available were included in this analysis (2003–2004, 2005–2006, 2007–2008, 2009–2010, 2011–2012, 2013–2014, and

2015–2016). The dietary and sample weighting data from the biennials were combined according to the NHANES analytic guidelines for combining surveys. From the combined data set, we estimated survey-design-weighted descriptive statistics for the population consumption per day.

Population statistics were estimated using the “survey” package version 3.36 (Lumley, 2004) in the R (version 3.6.1) environment for statistical computing (R Core Team, 2019) using the appropriate adjustment to sampling weights for combining biennials, then incorporating survey sampling units and strata from the survey design to ensure that sub-populations and areas were correctly represented. Descriptive statistics (mean, 90th percentile) were calculated for the subset of consumers of fava beans and for the entire population; these were broken down by age range and body-weight adjustment. Values were provided by individual food category and for total consumption of foods in all food categories.

2.4 Estimated Fava Bean Protein Daily Intake

Nuritas provided an assumption of 60% protein in the fava bean protein hydrolysate ingredient. Therefore, a factor of 0.6 was applied to all fava bean protein hydrolysate intake estimates to produce an estimate of fava bean protein intake.

3 Results

Tables 2 and 3 below, respectively, present the EDI for fava bean protein hydrolysate in grams per day and grams per kilogram body weight per day for the following age groups in the U.S. population: 2 years and older, 2 to 5 years, 6 to 18 years, 0 to 80 years, and 19 years and older. The “number of users” refers to the number of survey participants in a given age group who consumed a food item in given food category. The “percent users” is the percentage of fava bean protein hydrolysate users out of the total number of reliable survey participants (users and non-users) belonging to a given age group. Tables 4 and 5 present the estimates of fava bean protein intake in the same format.

Table 2. Estimated daily intake for fava bean protein hydrolysate (g/day)

Age Group (years)	Number of Users	Percent Users	EDI per user (g/day)		EDI per capita (g/day)	
			Mean	90th percentile	Mean	90th percentile
2–80	6866	12%	6.44	12.4	0.922	2.88
2–5	477	9%	4.21	7.17	0.439	0.586
6–18	2066	13%	5.64	11.2	0.860	2.88
0–80	7041	12%	6.44	12.4	0.906	2.88
19–80	4323	12%	6.76	13.3	0.971	2.88

Table 3. Estimated daily intake for fava bean protein hydrolysate (g/kg-bw/day)

Age Group (years)	Number of Users	Percent Users	EDI per user (g/day)		EDI per capita (g/day)	
			Mean	90th percentile	Mean	90th percentile
2–80	6866	12%	0.099	0.199	0.0142	0.0416
2–5	477	9%	0.258	0.436	0.0269	0.0320
6–18	2066	13%	0.116	0.217	0.0177	0.0592
0–80	7041	12%	0.104	0.205	0.0147	0.0410
19–80	4323	12%	0.0865	0.172	0.0124	0.0389

Table 4. Estimated daily intake for fava bean protein (g/day)

Age Group (years)	Number of Users	Percent Users	EDI per user (g/day)		EDI per capita (g/day)	
			Mean	90th percentile	Mean	90th percentile
2–80	6866	12%	3.86	7.45	0.553	1.73
2–5	477	9%	2.52	4.30	0.263	0.351
6–18	2066	13%	3.38	6.70	0.516	1.73
0–80	7041	12%	3.86	7.45	0.544	1.73
19–80	4323	12%	4.06	7.99	0.583	1.73

Table 5. Estimated daily intake for fava bean protein (g/kg-bw/day)

Age Group (years)	Number of Users	Percent Users	EDI per user (g/day)		EDI per capita (g/day)	
			Mean	90th percentile	Mean	90th percentile
2–80	6866	12%	0.0594	0.120	0.00851	0.0250
2–5	477	9%	0.155	0.262	0.0162	0.0192
6–18	2066	13%	0.0696	0.130	0.0106	0.0355
0–80	7041	12%	0.0625	0.123	0.00880	0.0246
19–80	4323	12%	0.0519	0.103	0.00746	0.0234

4 References

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APPENDIX A

List of Food Codes

Please see attached Excel document, "AppendixA.pdf."

Food Code	Long Description
11300100	Non-dairy milk, NFS
11511700	Nesquik, chocolate milk, ready to drink, low fat (1%), no sugar added
11511700	Nesquik, chocolate milk, ready to drink, low fat (1%), no sugar added
11513310	Chocolate milk, made from dry mix with non-dairy milk
11513375	Chocolate milk, made from reduced sugar mix with non-dairy milk
11513375	Chocolate milk, made from reduced sugar mix with non-dairy milk
11513375	Chocolate milk, made from reduced sugar mix with non-dairy milk
11513375	Chocolate milk, made from reduced sugar mix with non-dairy milk
11513385	Chocolate milk, made from dry mix with non-dairy milk (Nesquik)
11513385	Chocolate milk, made from dry mix with non-dairy milk (Nesquik)
11513385	Nesquik, chocolate milk, made from dry mix with non-dairy milk
11513385	Nesquik, chocolate milk, made from dry mix with non-dairy milk
11513395	Chocolate milk, made from no sugar added dry mix with non-dairy milk (Nesquik)
11513395	Chocolate milk, made from no sugar added dry mix with non-dairy milk (Nesquik)
11513395	Nesquik, chocolate milk, made from no sugar added dry mix with non-dairy milk
11513395	Nesquik, chocolate milk, made from no sugar added dry mix with non-dairy milk
11513750	Chocolate milk, made from syrup with non-dairy milk
11513805	Chocolate milk, made from light syrup with non-dairy milk
11513855	Chocolate milk, made from sugar free syrup with non-dairy milk
11513855	Chocolate milk, made from sugar free syrup with non-dairy milk
11513855	Chocolate milk, made from sugar free syrup with non-dairy milk
11513855	Chocolate milk, made from sugar free syrup with non-dairy milk
11553120	Fruit smoothie, with whole fruit and dairy, added protein
11613000	Instant breakfast, powder, sweetened with low calorie sweetener, milk added
11622010	Diet beverage, powder, reconstituted with skim milk
11623000	Meal supplement or replacement, commercially prepared, ready-to-drink
11631000	High calorie beverage, canned or powdered, reconstituted
11641000	Meal supplement or replacement, milk-based, high protein, liquid
11641020	Meal replacement or supplement, milk based, ready-to-drink
11651010	Meal replacement formula, Cambridge diet, reconstituted, all flavors
11830120	Cocoa, whey, and low calorie sweetener, fortified, dry mix, not reconstituted
11830800	Instant breakfast, powder, not reconstituted
11830810	Instant breakfast, powder, sweetened with low calorie sweetener, not reconstituted
11830850	High calorie milk beverage, powder, not reconstituted
11830900	Protein supplement, milk-based, powdered, not reconstituted

11830940 Meal replacement, high protein, milk based, fruit juice mixable formula, powdered, not reconstituted
 11830970 Meal replacement, protein type, milk-based, powdered, not reconstituted
 11830990 Nutrient supplement, milk-based, powdered, not reconstituted
 11831500 Nutrient supplement, milk-based, high protein, powdered, not reconstituted
 11832000 Meal replacement, protein type, milk- and soy-based, powdered, not reconstituted
 11835000 Meal replacement or nutritional supplement, Cambridge diet formula, powdered, nonfat milk solids base, dry, not reconstituted
 11835100 Meal replacement, Amway's Nutrilite brand Positrim Drink Mix, powdered nonfat dry milk-based, dry, not reconstituted
 11835150 Dynatrim, meal replacement, powder
 11836000 Protein supplement, milk-based, Muscle Milk, powdered, not reconstituted
 11836100 Protein supplement, milk-based, Muscle Milk Light, powdered, not reconstituted
 27564420 Frankfurter or hot dog sandwich, meatless, plain, on bun
 27564420 Frankfurter or hot dog sandwich, meatless, plain, on bun
 27564420 Frankfurter or hot dog sandwich, meatless, plain, on bun
 27564420 Frankfurter or hot dog sandwich, meatless, plain, on bun
 27564430 Frankfurter or hot dog sandwich, meatless, plain, on bread
 27564430 Frankfurter or hot dog sandwich, meatless, plain, on bread
 27564430 Frankfurter or hot dog sandwich, meatless, plain, on bread
 27564430 Frankfurter or hot dog sandwich, meatless, plain, on bread
 27564560 Frankfurter or hot dog sandwich, meatless, on bun, with meatless chili
 27564560 Frankfurter or hot dog sandwich, meatless, on bun, with meatless chili
 27564560 Frankfurter or hot dog sandwich, meatless, on bun, with meatless chili
 27564570 Frankfurter or hot dog sandwich, meatless, on bread, with meatless chili
 27564570 Frankfurter or hot dog sandwich, meatless, on bread, with meatless chili
 27564570 Frankfurter or hot dog sandwich, meatless, on bread, with meatless chili
 41430000 Protein powder, NFS
 41430010 Protein supplement, powdered
 41430200 Meal replacement or supplement, soy- and milk-base, powder, reconstituted with water
 41430310 Protein diet powder with soy and casein
 41435110 High protein bar, candy-like, soy and milk base
 41435120 Zone Perfect Classic Crunch nutrition bar
 41435700 South Beach Living High Protein Cereal Bar
 41435710 South Beach Living Meal Replacement Bar
 41436000 Nutritional supplement for people with diabetes, liquid
 41440010 Ensure liquid nutrition
 41440010 Ensure liquid nutrition

41440010 Meal replacement or supplement, liquid, soy-base, high protein
 41440010 Meal replacement or supplement, liquid, soy-base, high protein
 41440050 Ensure Plus liquid nutrition
 41440100 Meal replacement or supplement, liquid, soy-based
 41810200 Bacon strip, meatless
 41810250 Bacon bits, meatless
 41810250 Bacon bits, meatless
 41810400 Breakfast link, pattie, or slice, meatless
 41810400 Breakfast link, pattie, or slice, meatless
 41810400 Breakfast link, pattie, or slice, meatless
 41810400 Breakfast link, pattie, or slice, meatless
 41810600 Chicken, meatless, NFS
 41810600 Chicken, meatless, NFS
 41810600 Chicken, meatless, NFS
 41810600 Chicken, meatless, NFS
 41810610 Chicken, meatless, breaded, fried
 41810610 Chicken, meatless, breaded, fried
 41810610 Chicken, meatless, breaded, fried
 41810610 Chicken, meatless, breaded, fried
 41811200 Fish stick, meatless
 41811400 Frankfurter or hot dog, meatless
 41811600 Luncheon slice, meatless-beef, chicken, salami or turkey
 41811600 Luncheon slice, meatless-beef, chicken, salami or turkey
 41811600 Luncheon slice, meatless-beef, chicken, salami or turkey
 41811600 Luncheon slice, meatless-beef, chicken, salami or turkey
 41811800 Meatball, meatless
 41811890 Vegetarian burger or patty, meatless, no bun
 41811910 Vegetable burger or patty, meatless, no bun
 52202500 Cornbread, made with egg substitute, made from home recipe
 53231400 Cookie, multigrain, high fiber
 53260030 Cookie, dietetic, chocolate chip
 53260030 Cookie, dietetic, chocolate chip
 53260200 Cookie, dietetic, oatmeal with raisins
 53260200 Cookie, dietetic, oatmeal with raisins
 53260300 Cookie, dietetic, sandwich type
 53260300 Cookie, dietetic, sandwich type
 53260400 Cookie, dietetic, sugar or plain
 53260400 Cookie, dietetic, sugar or plain

53541200 Meal replacement bar
 53541200 Meal replacement bar
 53541200 Meal replacement bar
 53541200 Meal replacement bar
 53544450 PowerBar (fortified high energy bar)
 53720100 Nutrition bar (Balance Original Bar)
 53720100 Nutrition bar (Balance Original Bar)
 53720200 Nutrition bar (Clif Bar)
 53720200 Nutrition bar (Clif Bar)
 53720210 Nutrition bar (Clif Kids Organic Zbar)
 53720210 Nutrition bar (Clif Kids Organic Zbar)
 53720300 Nutrition bar (PowerBar)
 53720300 Nutrition bar (PowerBar)
 53720400 Nutrition bar (Slim Fast Original Meal Bar)
 53720400 Nutrition bar (Slim Fast Original Meal Bar)
 53720500 Nutrition bar (Snickers Marathon Protein Bar)
 53720500 Nutrition bar (Snickers Marathon Protein Bar)
 53720500 Snickers Marathon Protein bar
 53720500 Snickers Marathon Protein bar
 53720510 Snickers Marathon Energy bar
 53720600 Nutrition bar (South Beach Living Meal Bar)
 53720600 Nutrition bar (South Beach Living Meal Bar)
 53720610 Nutrition bar (South Beach Living High Protein Bar)
 53720610 Nutrition bar (South Beach Living High Protein Bar)
 53720610 South Beach Living High Protein Bar
 53720610 South Beach Living High Protein Bar
 53720700 Nutrition bar (Tiger's Milk)
 53720700 Nutrition bar (Tiger's Milk)
 53720800 Nutrition bar (Zone Perfect Classic Crunch)
 53720800 Nutrition bar (Zone Perfect Classic Crunch)
 53720800 Zone Perfect Classic Crunch nutrition bar
 53720800 Zone Perfect Classic Crunch nutrition bar
 53729000 Nutrition bar or meal replacement bar, NFS
 54103000 Crackers, breakfast biscuit
 56201065 Grits, regular or quick, made with non-dairy milk, NS as to fat
 56201065 Grits, regular or quick, made with non-dairy milk, NS as to fat
 56201065 Grits, regular or quick, made with non-dairy milk, NS as to fat added in cooking
 56201065 Grits, regular or quick, made with non-dairy milk, NS as to fat added in cooking

56201066 Grits, regular or quick, made with non-dairy milk, fat not added in cooking
56201066 Grits, regular or quick, made with non-dairy milk, fat not added in cooking
56201066 Grits, regular or quick, made with non-dairy milk, no added fat
56201066 Grits, regular or quick, made with non-dairy milk, no added fat
56201067 Grits, regular or quick, made with non-dairy milk, fat added
56201067 Grits, regular or quick, made with non-dairy milk, fat added
56201067 Grits, regular or quick, made with non-dairy milk, fat added in cooking
56201067 Grits, regular or quick, made with non-dairy milk, fat added in cooking
56201350 Grits, instant, made with non-dairy milk, NS as to fat
56201350 Grits, instant, made with non-dairy milk, NS as to fat
56201350 Grits, instant, made with non-dairy milk, NS as to fat added in cooking
56201350 Grits, instant, made with non-dairy milk, NS as to fat added in cooking
56201355 Grits, instant, made with non-dairy milk, fat not added in cooking
56201355 Grits, instant, made with non-dairy milk, fat not added in cooking
56201355 Grits, instant, made with non-dairy milk, no added fat
56201355 Grits, instant, made with non-dairy milk, no added fat
56201360 Grits, instant, made with non-dairy milk, fat added
56201360 Grits, instant, made with non-dairy milk, fat added
56201360 Grits, instant, made with non-dairy milk, fat added in cooking
56201360 Grits, instant, made with non-dairy milk, fat added in cooking
56203075 Oatmeal, regular or quick, made with non-dairy milk, NS as to fat
56203075 Oatmeal, regular or quick, made with non-dairy milk, NS as to fat
56203075 Oatmeal, regular or quick, made with non-dairy milk, NS as to fat added in cooking
56203075 Oatmeal, regular or quick, made with non-dairy milk, NS as to fat added in cooking
56203076 Oatmeal, regular or quick, made with non-dairy milk, fat not added in cooking
56203076 Oatmeal, regular or quick, made with non-dairy milk, fat not added in cooking
56203076 Oatmeal, regular or quick, made with non-dairy milk, no added fat
56203076 Oatmeal, regular or quick, made with non-dairy milk, no added fat
56203077 Oatmeal, regular or quick, made with non-dairy milk, fat added
56203077 Oatmeal, regular or quick, made with non-dairy milk, fat added
56203077 Oatmeal, regular or quick, made with non-dairy milk, fat added in cooking
56203077 Oatmeal, regular or quick, made with non-dairy milk, fat added in cooking
56203105 Oatmeal, instant, plain, made with non-dairy milk, NS as to fat
56203105 Oatmeal, instant, plain, made with non-dairy milk, NS as to fat
56203105 Oatmeal, instant, plain, made with non-dairy milk, NS as to fat added in cooking
56203105 Oatmeal, instant, plain, made with non-dairy milk, NS as to fat added in cooking
56203106 Oatmeal, instant, plain, made with non-dairy milk, fat not added in cooking
56203106 Oatmeal, instant, plain, made with non-dairy milk, fat not added in cooking

56203106 Oatmeal, instant, plain, made with non-dairy milk, no added fat
56203106 Oatmeal, instant, plain, made with non-dairy milk, no added fat
56203107 Oatmeal, instant, plain, made with non-dairy milk, fat added
56203107 Oatmeal, instant, plain, made with non-dairy milk, fat added
56203107 Oatmeal, instant, plain, made with non-dairy milk, fat added in cooking
56203107 Oatmeal, instant, plain, made with non-dairy milk, fat added in cooking
56207025 Cream of wheat, regular or quick, made with non-dairy milk, NS as to fat
56207025 Cream of wheat, regular or quick, made with non-dairy milk, NS as to fat
56207025 Cream of wheat, regular or quick, made with non-dairy milk, NS as to fat added in cooking
56207025 Cream of wheat, regular or quick, made with non-dairy milk, NS as to fat added in cooking
56207026 Cream of wheat, regular or quick, made with non-dairy milk, fat not added in cooking
56207026 Cream of wheat, regular or quick, made with non-dairy milk, fat not added in cooking
56207026 Cream of wheat, regular or quick, made with non-dairy milk, no added fat
56207026 Cream of wheat, regular or quick, made with non-dairy milk, no added fat
56207027 Cream of wheat, regular or quick, made with non-dairy milk, fat added
56207027 Cream of wheat, regular or quick, made with non-dairy milk, fat added
56207027 Cream of wheat, regular or quick, made with non-dairy milk, fat added in cooking
56207027 Cream of wheat, regular or quick, made with non-dairy milk, fat added in cooking
56207101 Cream of wheat, instant, made with non-dairy milk, NS as to fat
56207101 Cream of wheat, instant, made with non-dairy milk, NS as to fat
56207101 Cream of wheat, instant, made with non-dairy milk, NS as to fat added in cooking
56207101 Cream of wheat, instant, made with non-dairy milk, NS as to fat added in cooking
56207102 Cream of wheat, instant, made with non-dairy milk, fat not added in cooking
56207102 Cream of wheat, instant, made with non-dairy milk, fat not added in cooking
56207102 Cream of wheat, instant, made with non-dairy milk, no added fat
56207102 Cream of wheat, instant, made with non-dairy milk, no added fat
56207103 Cream of wheat, instant, made with non-dairy milk, fat added
56207103 Cream of wheat, instant, made with non-dairy milk, fat added
56207103 Cream of wheat, instant, made with non-dairy milk, fat added in cooking
56207103 Cream of wheat, instant, made with non-dairy milk, fat added in cooking
57316385 Cereal (General Mills Cheerios Protein)
91708100 Fruit snack candy, with added vitamin C
91708100 Fruit snack candy, with added vitamin C
91708100 Fruit snack candy, with added vitamin C
91708100 Fruit snacks candy, with high vitamin C

91708100 Fruit snacks candy, with high vitamin C
91708100 Fruit snacks candy, with high vitamin C
91708100 Fruit snacks candy, with high vitamin C
91708100 Fruit snacks candy, with high vitamin C
91708100 Fruit snacks candy, with high vitamin C
91708150 Yogurt covered fruit snacks candy, with added vitamin C
91708150 Yogurt covered fruit snacks candy, with added vitamin C
91708150 Yogurt covered fruit snacks candy, with added vitamin C
91708150 Yogurt covered fruit snacks candy, with added vitamin C
91708160 Yogurt covered fruit snacks candy rolls, with high vitamin C
91708160 Yogurt covered fruit snacks candy rolls, with high vitamin C
91708160 Yogurt covered fruit snacks candy rolls, with high vitamin C
91708160 Yogurt covered fruit snacks candy rolls, with high vitamin C
91780010 Snickers Marathon Energy bar
91781010 Snickers Marathon Protein bar
92560100 Gatorade Thirst Quencher sports drink
92560200 Powerade sports drink
92565000 Fruit-flavored sports drink or thirst quencher beverage, low calorie
92900300 Sports drink, dry concentrate, not reconstituted
92900300 Sports drink, dry concentrate, not reconstituted
95101000 Boost, nutritional drink, ready-to-drink
95101000 Boost, nutritional drink, ready-to-drink
95101000 Nutritional drink or shake, ready-to-drink (Boost)
95101000 Nutritional drink or shake, ready-to-drink (Boost)
95101010 Boost Plus, nutritional drink, ready-to-drink
95101010 Boost Plus, nutritional drink, ready-to-drink
95101010 Nutritional drink or shake, ready-to-drink (Boost Plus)
95101010 Nutritional drink or shake, ready-to-drink (Boost Plus)
95102000 Carnation Instant Breakfast, nutritional drink, regular, ready-to-drink
95102000 Carnation Instant Breakfast, nutritional drink, regular, ready-to-drink
95102000 Nutritional drink or shake, ready-to-drink (Carnation Instant Breakfast)
95102000 Nutritional drink or shake, ready-to-drink (Carnation Instant Breakfast)
95102010 Carnation Instant Breakfast, nutritional drink, sugar free, ready-to-drink
95103000 Ensure, nutritional shake, ready-to-drink
95103000 Ensure, nutritional shake, ready-to-drink
95103000 Nutritional drink or shake, ready-to-drink (Ensure)
95103000 Nutritional drink or shake, ready-to-drink (Ensure)
95103010 Ensure Plus, nutritional shake, ready-to-drink

95103010 Ensure Plus, nutritional shake, ready-to-drink
95103010 Nutritional drink or shake, ready-to-drink (Ensure Plus)
95103010 Nutritional drink or shake, ready-to-drink (Ensure Plus)
95104000 Glucerna, nutritional shake, ready-to-drink
95104000 Glucerna, nutritional shake, ready-to-drink
95104000 Nutritional drink or shake, ready-to-drink, sugar free (Glucerna)
95104000 Nutritional drink or shake, ready-to-drink, sugar free (Glucerna)
95105000 Kellogg's Special K Protein Shake
95105000 Kellogg's Special K Protein Shake
95105000 Nutritional drink or shake, ready-to-drink (Kellogg's Special K Protein)
95105000 Nutritional drink or shake, ready-to-drink (Kellogg's Special K Protein)
95106000 Muscle Milk, ready-to-drink
95106000 Muscle Milk, ready-to-drink
95106000 Nutritional drink or shake, ready-to-drink (Muscle Milk)
95106000 Nutritional drink or shake, ready-to-drink (Muscle Milk)
95106010 Muscle Milk, light, ready-to-drink
95106010 Muscle Milk, light, ready-to-drink
95106010 Nutritional drink or shake, ready-to-drink, light (Muscle Milk)
95106010 Nutritional drink or shake, ready-to-drink, light (Muscle Milk)
95110000 Nutritional drink or shake, ready-to-drink (Slim Fast)
95110000 Nutritional drink or shake, ready-to-drink (Slim Fast)
95110000 Slim Fast Shake, meal replacement, regular, ready-to-drink
95110000 Slim Fast Shake, meal replacement, regular, ready-to-drink
95110010 Nutritional drink or shake, ready-to-drink, sugar free (Slim Fast)
95110010 Nutritional drink or shake, ready-to-drink, sugar free (Slim Fast)
95110010 Slim Fast Shake, meal replacement, sugar free, ready-to-drink
95110010 Slim Fast Shake, meal replacement, sugar free, ready-to-drink
95110020 Nutritional drink or shake, high protein, ready-to-drink (Slim Fast)
95110020 Nutritional drink or shake, high protein, ready-to-drink (Slim Fast)
95110020 Slim Fast Shake, meal replacement, high protein, ready-to-drink
95110020 Slim Fast Shake, meal replacement, high protein, ready-to-drink
95120000 Nutritional drink or meal replacement, ready-to-drink, NFS
95120000 Nutritional drink or meal replacement, ready-to-drink, NFS
95120000 Nutritional drink or shake, ready-to-drink, NFS
95120000 Nutritional drink or shake, ready-to-drink, NFS
95120010 Nutritional drink or meal replacement, high protein, ready-to-drink, NFS
95120010 Nutritional drink or meal replacement, high protein, ready-to-drink, NFS
95120010 Nutritional drink or shake, high protein, ready-to-drink, NFS

95120010 Nutritional drink or shake, high protein, ready-to-drink, NFS
 95120020 Nutritional drink or meal replacement, high protein, light, ready-to-drink, NFS
 95120020 Nutritional drink or meal replacement, high protein, light, ready-to-drink, NFS
 95120020 Nutritional drink or shake, high protein, light, ready-to-drink, NFS
 95120020 Nutritional drink or shake, high protein, light, ready-to-drink, NFS
 95120050 Nutritional drink or meal replacement, liquid, soy-based
 95120050 Nutritional drink or meal replacement, liquid, soy-based
 95120050 Nutritional drink or shake, liquid, soy-based
 95120050 Nutritional drink or shake, liquid, soy-based
 95201000 Carnation Instant Breakfast, nutritional drink mix, regular, powder
 95201000 Carnation Instant Breakfast, nutritional drink mix, regular, powder
 95201000 Nutritional powder mix (Carnation Instant Breakfast)
 95201000 Nutritional powder mix (Carnation Instant Breakfast)
 95201010 Carnation Instant Breakfast, nutritional drink mix, sugar free, powder
 95201010 Carnation Instant Breakfast, nutritional drink mix, sugar free, powder
 95201010 Nutritional powder mix, sugar free (Carnation Instant Breakfast)
 95201010 Nutritional powder mix, sugar free (Carnation Instant Breakfast)
 95201200 EAS Whey Protein Powder
 95201200 EAS Whey Protein Powder
 95201200 Nutritional powder mix (EAS Whey Protein Powder)
 95201200 Nutritional powder mix (EAS Whey Protein Powder)
 95201300 EAS Soy Protein Powder
 95201300 EAS Soy Protein Powder
 95201300 Nutritional powder mix (EAS Soy Protein Powder)
 95201300 Nutritional powder mix (EAS Soy Protein Powder)
 95201500 Herbalife, nutritional shake mix, high protein, powder
 95201500 Herbalife, nutritional shake mix, high protein, powder
 95201500 Nutritional powder mix, high protein (Herbalife)
 95201500 Nutritional powder mix, high protein (Herbalife)
 95201600 Isopure protein powder
 95201600 Isopure protein powder
 95201600 Nutritional powder mix (Isopure)
 95201600 Nutritional powder mix (Isopure)
 95202000 Muscle Milk, regular, powder
 95202000 Muscle Milk, regular, powder
 95202000 Nutritional powder mix (Muscle Milk)
 95202000 Nutritional powder mix (Muscle Milk)
 95202010 Muscle Milk, light, powder

95202010 Muscle Milk, light, powder
95202010 Nutritional powder mix, light (Muscle Milk)
95202010 Nutritional powder mix, light (Muscle Milk)
95210000 Nutritional powder mix (Slim Fast)
95210000 Nutritional powder mix (Slim Fast)
95210000 Slim Fast Shake Mix, powder
95210000 Slim Fast Shake Mix, powder
95210010 Nutritional powder mix, sugar free (Slim Fast)
95210010 Nutritional powder mix, sugar free (Slim Fast)
95210010 Slim Fast Shake Mix, sugar free, powder
95210010 Slim Fast Shake Mix, sugar free, powder
95210020 Nutritional powder mix, high protein (Slim Fast)
95210020 Nutritional powder mix, high protein (Slim Fast)
95210020 Slim Fast Shake Mix, high protein, powder
95210020 Slim Fast Shake Mix, high protein, powder
95220000 Nutritional drink mix or meal replacement, powder, NFS
95220000 Nutritional drink mix or meal replacement, powder, NFS
95220000 Nutritional powder mix, NFS
95220000 Nutritional powder mix, NFS
95220010 Nutritional drink mix or meal replacement, high protein, powder, NFS
95220010 Nutritional drink mix or meal replacement, high protein, powder, NFS
95220010 Nutritional powder mix, high protein, NFS
95220010 Nutritional powder mix, high protein, NFS
95230000 Nutritional powder mix, whey based, NFS
95230000 Nutritional powder mix, whey based, NFS
95230000 Protein powder, whey based, NFS
95230000 Protein powder, whey based, NFS
95230010 Nutritional powder mix, protein, soy based, NFS
95230010 Nutritional powder mix, protein, soy based, NFS
95230010 Protein powder, soy based, NFS
95230010 Protein powder, soy based, NFS
95230020 Nutritional powder mix, protein, light, NFS
95230020 Nutritional powder mix, protein, light, NFS
95230020 Protein powder, light, NFS
95230020 Protein powder, light, NFS
95230030 Protein powder, NFS
95230030 Protein powder, NFS
95230030 Nutritional powder mix, protein, NFS

95230030 Nutritional powder mix, protein, NFS
95320200 Gatorade G sports drink
95320200 Gatorade G sports drink
95320200 Gatorade G sports drink
95320200 Gatorade Thirst Quencher sports drink
95320200 Gatorade Thirst Quencher sports drink
95320200 Gatorade Thirst Quencher sports drink
95320200 Sports drink (Gatorade G)
95320200 Sports drink (Gatorade G)
95320200 Sports drink (Gatorade G)
95320500 Powerade sports drink
95320500 Powerade sports drink
95320500 Sports drink (Powerade)
95320500 Sports drink (Powerade)
95321000 Sports drink, NFS
95321000 Sports drink, NFS
95322200 Gatorade G2 sports drink, low calorie
95322200 Gatorade G2 sports drink, low calorie
95322200 Gatorade G2 sports drink, low calorie
95322200 Gatorade G2 Thirst Quencher sports drink, low calorie
95322200 Gatorade G2 Thirst Quencher sports drink, low calorie
95322200 Gatorade G2 Thirst Quencher sports drink, low calorie
95322200 Sports drink, low calorie (Gatorade G2)
95322200 Sports drink, low calorie (Gatorade G2)
95322200 Sports drink, low calorie (Gatorade G2)
95322500 Powerade Zero sports drink, low calorie
95322500 Powerade Zero sports drink, low calorie
95322500 Sports drink, low calorie (Powerade Zero)
95322500 Sports drink, low calorie (Powerade Zero)
95323000 Fruit-flavored sports drink or thirst quencher beverage, low calorie
95323000 Fruit-flavored sports drink or thirst quencher beverage, low calorie
95323000 Sports drink, low calorie
95323000 Sports drink, low calorie
95330100 Fluid replacement, electrolyte solution

APPENDIX B

List of Food Code Proportions

Please see attached Excel document, "AppendixB.pdf."

Food_Code ID	Category	Description	Use_Level (g/serving)	Use Level (%)
95330100	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Fluid replacement, electrolyte solution	2.4	8%
95323000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Fruit-flavored sports drink or thirst quencher beverage, low calorie	2.4	1%
95323000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Sports drink, low calorie	2.4	1%
95322500	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Powerade Zero sports drink, low calorie	2.4	1%
95322500	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Sports drink, low calorie (Powerade Zero)	2.4	1%
95322200	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Gatorade G2 sports drink, low calorie	2.4	1%
95322200	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Gatorade G2 Thirst Quencher sports drink, low calorie	2.4	1%
95322200	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Sports drink, low calorie (Gatorade G2)	2.4	1%
95321000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Sports drink, NFS	2.4	1%

95320500	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Powerade sports drink	2.4	1%
95320500	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Sports drink (Powerade)	2.4	1%
95320200	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Gatorade G sports drink	2.4	1%
95320200	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Gatorade Thirst Quencher sports drink	2.4	1%
95320200	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Sports drink (Gatorade G)	2.4	1%
95230030	Dry blend powders - nutritionally fortified beverage mixes	Protein powder, NFS	2.4	8%
95230030	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, protein, NFS	2.4	8%
95230020	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, protein, light, NFS	2.4	30%
95230020	Dry blend powders - nutritionally fortified beverage mixes	Protein powder, light, NFS	2.4	30%
95230010	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, protein, soy based, NFS	2.4	7%
95230010	Dry blend powders - nutritionally fortified beverage mixes	Protein powder, soy based, NFS	2.4	7%
95230000	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, whey based, NFS	2.4	9%
95230000	Dry blend powders - nutritionally fortified beverage mixes	Protein powder, whey based, NFS	2.4	9%

95220010	Dry blend powders - nutritionally fortified beverage mixes	Nutritional drink mix or meal replacement, high protein, powder, NFS	2.4	5%
95220010	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, high protein, NFS	2.4	5%
95220000	Dry blend powders - nutritionally fortified beverage mixes	Nutritional drink mix or meal replacement, powder, NFS	2.4	7%
95220000	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, NFS	2.4	7%
95210020	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, high protein (Slim Fast)	2.4	9%
95210020	Dry blend powders - nutritionally fortified beverage mixes	Slim Fast Shake Mix, high protein, powder	2.4	9%
95210010	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, sugar free (Slim Fast)	2.4	17%
95210010	Dry blend powders - nutritionally fortified beverage mixes	Slim Fast Shake Mix, sugar free, powder	2.4	17%
95210000	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix (Slim Fast)	2.4	9%
95210000	Dry blend powders - nutritionally fortified beverage mixes	Slim Fast Shake Mix, powder	2.4	9%
95202010	Dry blend powders - nutritionally fortified beverage mixes	Muscle Milk, light, powder	2.4	10%
95202010	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, light (Muscle Milk)	2.4	10%
95202000	Dry blend powders - nutritionally fortified beverage mixes	Muscle Milk, regular, powder	2.4	7%
95202000	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix (Muscle Milk)	2.4	7%
95201600	Dry blend powders - nutritionally fortified beverage mixes	Isopure protein powder	2.4	9%

95201600	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix (Isopure)	2.4	9%
95201500	Dry blend powders - nutritionally fortified beverage mixes	Herbalife, nutritional shake mix, high protein, powder	2.4	19%
95201500	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, high protein (Herbalife)	2.4	19%
95201300	Dry blend powders - nutritionally fortified beverage mixes	EAS Soy Protein Powder	2.4	5%
95201300	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix (EAS Soy Protein Powder)	2.4	5%
95201200	Dry blend powders - nutritionally fortified beverage mixes	EAS Whey Protein Powder	2.4	12%
95201200	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix (EAS Whey Protein Powder)	2.4	12%
95201010	Dry blend powders - nutritionally fortified beverage mixes	Carnation Instant Breakfast, nutritional drink mix, sugar free, powder	2.4	12%
95201010	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix, sugar free (Carnation Instant Breakfast)	2.4	12%
95201000	Dry blend powders - nutritionally fortified beverage mixes	Carnation Instant Breakfast, nutritional drink mix, regular, powder	2.4	6%
95201000	Dry blend powders - nutritionally fortified beverage mixes	Nutritional powder mix (Carnation Instant Breakfast)	2.4	6%
95120050	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or meal replacement, liquid, soy-based	2.4	1%
95120050	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, liquid, soy-based	2.4	1%
95120020	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or meal replacement, high protein, light, ready-to-drink, NFS	2.4	1%

95120020	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, high protein, light, ready-to-drink, NFS	2.4	1%
95120010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or meal replacement, high protein, ready-to-drink, NFS	2.4	0%
95120010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, high protein, ready-to-drink, NFS	2.4	0%
95120000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or meal replacement, ready-to-drink, NFS	2.4	1%
95120000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink, NFS	2.4	1%
95110020	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, high protein, ready-to-drink (Slim Fast)	2.4	1%
95110020	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Slim Fast Shake, meal replacement, high protein, ready-to-drink	2.4	1%
95110010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink, sugar free (Slim Fast)	2.4	1%
95110010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Slim Fast Shake, meal replacement, sugar free, ready-to-drink	2.4	1%
95110000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink (Slim Fast)	2.4	1%

95110000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Slim Fast Shake, meal replacement, regular, ready-to-drink	2.4	1%
95106010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Muscle Milk, light, ready-to-drink	2.4	1%
95106010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink, light (Muscle Milk)	2.4	1%
95106000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Muscle Milk, ready-to-drink	2.4	1%
95106000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink (Muscle Milk)	2.4	1%
95105000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Kellogg's Special K Protein Shake	2.4	1%
95105000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink (Kellogg's Special K Protein)	2.4	1%
95104000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Glucerna, nutritional shake, ready-to-drink	2.4	1%
95104000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink, sugar free (Glucerna)	2.4	1%
95103010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Ensure Plus, nutritional shake, ready-to-drink	2.4	1%

95103010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink (Ensure Plus)	2.4	1%
95103000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Ensure, nutritional shake, ready-to-drink	2.4	1%
95103000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink (Ensure)	2.4	1%
95102010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Carnation Instant Breakfast, nutritional drink, sugar free, ready-to-drink	2.4	1%
95102000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Carnation Instant Breakfast, nutritional drink, regular, ready-to-drink	2.4	1%
95102000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink (Carnation Instant Breakfast)	2.4	1%
95101010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Boost Plus, nutritional drink, ready-to-drink	2.4	1%
95101010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink (Boost Plus)	2.4	1%
95101000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Boost, nutritional drink, ready-to-drink	2.4	1%
95101000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional drink or shake, ready-to-drink (Boost)	2.4	1%

92900300	Dry blend powders - nutritionally fortified beverage mixes	Sports drink, dry concentrate, not reconstituted	2.4	7%
92565000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Fruit-flavored sports drink or thirst quencher beverage, low calorie	2.4	1%
92560200	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Powerade sports drink	2.4	1%
92560100	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Gatorade Thirst Quencher sports drink	2.4	1%
91781010	Nutritionally fortified snack foods – protein bars, energy bars	Snickers Marathon Protein bar	2.4	3%
91780010	Nutritionally fortified snack foods – protein bars, energy bars	Snickers Marathon Energy bar	2.4	5%
91708160	Nutritionally fortified confectionary – hard candies, soft gummies	Yogurt covered fruit snacks candy rolls, with high vitamin C	2.4	10%
91708160	Nutritionally fortified confectionary – hard candies, soft gummies	Yogurt covered fruit snacks candy rolls, with high vitamin C	2.4	10%
91708150	Nutritionally fortified confectionary – hard candies, soft gummies	Yogurt covered fruit snacks candy, with added vitamin C	2.4	12%
91708150	Nutritionally fortified confectionary – hard candies, soft gummies	Yogurt covered fruit snacks candy, with added vitamin C	2.4	12%
91708100	Nutritionally fortified confectionary – hard candies, soft gummies	Fruit snack candy, with added vitamin C	2.4	9%

91708100	Nutritionally fortified confectionary – hard candies, soft gummies	Fruit snacks candy, with high vitamin C	2.4	9%
91708100	Nutritionally fortified confectionary – hard candies, soft gummies	Fruit snacks candy, with high vitamin C	2.4	9%
57316385	Baked goods – nutritionally fortified cookies, biscuits, cereal bars	Cereal (General Mills Cheerios Protein)	2.4	9%
56207103	Dairy alternatives, Dairy analogs	Cream of wheat, instant, made with non-dairy milk, fat added	2.4	1%
56207103	Dairy alternatives, Dairy analogs	Cream of wheat, instant, made with non-dairy milk, fat added in cooking	2.4	1%
56207102	Dairy alternatives, Dairy analogs	Cream of wheat, instant, made with non-dairy milk, fat not added in cooking	2.4	1%
56207102	Dairy alternatives, Dairy analogs	Cream of wheat, instant, made with non-dairy milk, no added fat	2.4	1%
56207101	Dairy alternatives, Dairy analogs	Cream of wheat, instant, made with non-dairy milk, NS as to fat	2.4	1%
56207101	Dairy alternatives, Dairy analogs	Cream of wheat, instant, made with non-dairy milk, NS as to fat added in cooking	2.4	1%
56207027	Dairy alternatives, Dairy analogs	Cream of wheat, regular or quick, made with non-dairy milk, fat added	2.4	1%
56207027	Dairy alternatives, Dairy analogs	Cream of wheat, regular or quick, made with non-dairy milk, fat added in cooking	2.4	1%
56207026	Dairy alternatives, Dairy analogs	Cream of wheat, regular or quick, made with non-dairy milk, fat not added in cooking	2.4	1%

56207026	Dairy alternatives, Dairy analogs	Cream of wheat, regular or quick, made with non-dairy milk, no added fat	2.4	1%
56207025	Dairy alternatives, Dairy analogs	Cream of wheat, regular or quick, made with non-dairy milk, NS as to fat	2.4	1%
56207025	Dairy alternatives, Dairy analogs	Cream of wheat, regular or quick, made with non-dairy milk, NS as to fat added in cooking	2.4	1%
56203107	Dairy alternatives, Dairy analogs	Oatmeal, instant, plain, made with non-dairy milk, fat added	2.4	1%
56203107	Dairy alternatives, Dairy analogs	Oatmeal, instant, plain, made with non-dairy milk, fat added in cooking	2.4	1%
56203106	Dairy alternatives, Dairy analogs	Oatmeal, instant, plain, made with non-dairy milk, fat not added in cooking	2.4	1%
56203106	Dairy alternatives, Dairy analogs	Oatmeal, instant, plain, made with non-dairy milk, no added fat	2.4	1%
56203105	Dairy alternatives, Dairy analogs	Oatmeal, instant, plain, made with non-dairy milk, NS as to fat	2.4	1%
56203105	Dairy alternatives, Dairy analogs	Oatmeal, instant, plain, made with non-dairy milk, NS as to fat added in cooking	2.4	1%
56203077	Dairy alternatives, Dairy analogs	Oatmeal, regular or quick, made with non-dairy milk, fat added	2.4	1%
56203077	Dairy alternatives, Dairy analogs	Oatmeal, regular or quick, made with non-dairy milk, fat added in cooking	2.4	1%
56203076	Dairy alternatives, Dairy analogs	Oatmeal, regular or quick, made with non-dairy milk, fat not added in cooking	2.4	1%
56203076	Dairy alternatives, Dairy analogs	Oatmeal, regular or quick, made with non-dairy milk, no added fat	2.4	1%
56203075	Dairy alternatives, Dairy analogs	Oatmeal, regular or quick, made with non-dairy milk, NS as to fat	2.4	1%

56203075	Dairy alternatives, Dairy analogs	Oatmeal, regular or quick, made with non-dairy milk, NS as to fat added in cooking	2.4	1%
56201360	Dairy alternatives, Dairy analogs	Grits, instant, made with non-dairy milk, fat added	2.4	1%
56201360	Dairy alternatives, Dairy analogs	Grits, instant, made with non-dairy milk, fat added in cooking	2.4	1%
56201355	Dairy alternatives, Dairy analogs	Grits, instant, made with non-dairy milk, fat not added in cooking	2.4	1%
56201355	Dairy alternatives, Dairy analogs	Grits, instant, made with non-dairy milk, no added fat	2.4	1%
56201350	Dairy alternatives, Dairy analogs	Grits, instant, made with non-dairy milk, NS as to fat	2.4	1%
56201350	Dairy alternatives, Dairy analogs	Grits, instant, made with non-dairy milk, NS as to fat added in cooking	2.4	1%
56201067	Dairy alternatives, Dairy analogs	Grits, regular or quick, made with non-dairy milk, fat added	2.4	1%
56201067	Dairy alternatives, Dairy analogs	Grits, regular or quick, made with non-dairy milk, fat added in cooking	2.4	1%
56201066	Dairy alternatives, Dairy analogs	Grits, regular or quick, made with non-dairy milk, fat not added in cooking	2.4	1%
56201066	Dairy alternatives, Dairy analogs	Grits, regular or quick, made with non-dairy milk, no added fat	2.4	1%
56201065	Dairy alternatives, Dairy analogs	Grits, regular or quick, made with non-dairy milk, NS as to fat	2.4	1%
56201065	Dairy alternatives, Dairy analogs	Grits, regular or quick, made with non-dairy milk, NS as to fat added in cooking	2.4	1%
54103000	Baked goods – nutritionally fortified cookies, biscuits, cereal bars	Crackers, breakfast biscuit	2.4	16%

53729000	Meal replacement drinks/Meal Replacement Bars	Nutrition bar or meal replacement bar, NFS	2.4	7%
53720800	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (Zone Perfect Classic Crunch)	2.4	5%
53720800	Nutritionally fortified snack foods – protein bars, energy bars	Zone Perfect Classic Crunch nutrition bar	2.4	5%
53720700	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (Tiger's Milk)	2.4	7%
53720610	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (South Beach Living High Protein Bar)	2.4	7%
53720610	Nutritionally fortified snack foods – protein bars, energy bars	South Beach Living High Protein Bar	2.4	7%
53720600	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (South Beach Living Meal Bar)	2.4	5%
53720510	Nutritionally fortified snack foods – protein bars, energy bars	Snickers Marathon Energy bar	2.4	5%
53720500	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (Snickers Marathon Protein Bar)	2.4	3%
53720500	Nutritionally fortified snack foods – protein bars, energy bars	Snickers Marathon Protein bar	2.4	3%
53720400	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (Slim Fast Original Meal Bar)	2.4	4%
53720300	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (PowerBar)	2.4	4%
53720210	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (Clif Kids Organic Zbar)	2.4	7%

53720200	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (Clif Bar)	2.4	4%
53720100	Nutritionally fortified snack foods – protein bars, energy bars	Nutrition bar (Balance Original Bar)	2.4	5%
53544450	Nutritionally fortified snack foods – protein bars, energy bars	PowerBar (fortified high energy bar)	2.4	4%
53541200	Meal replacement drinks/Meal Replacement Bars	Meal replacement bar	2.4	4%
53541200	Meal replacement drinks/Meal Replacement Bars	Meal replacement bar	2.4	4%
53260400	Baked goods – nutritionally fortified cookies, biscuits, cereal bars	Cookie, dietetic, sugar or plain	2.4	4%
53260300	Baked goods – nutritionally fortified cookies, biscuits, cereal bars	Cookie, dietetic, sandwich type	2.4	7%
53260200	Baked goods – nutritionally fortified cookies, biscuits, cereal bars	Cookie, dietetic, oatmeal with raisins	2.4	4%
53260030	Baked goods – nutritionally fortified cookies, biscuits, cereal bars	Cookie, dietetic, chocolate chip	2.4	4%
53231400	Baked goods – nutritionally fortified cookies, biscuits, cereal bars	Cookie, multigrain, high fiber	2.4	4%
52202500	Dairy alternatives, Dairy analogs	Cornbread, made with egg substitute, made from home recipe	2.4	1%
41811910	Plant-based burgers	Vegetable burger or patty, meatless, no bun	2.4	3%

41811890	Plant-based burgers	Vegetarian burger or patty, meatless, no bun	2.4	3%
41811800	Plant-based burgers	Meatball, meatless	2.4	4%
41811600	Plant-based burgers	Luncheon slice, meatless-beef, chicken, salami or turkey	2.4	4%
41811600	Plant-based burgers	Luncheon slice, meatless-beef, chicken, salami or turkey	2.4	4%
41811400	Plant-based burgers	Frankfurter or hot dog, meatless	2.4	3%
41811200	Plant-based burgers	Fish stick, meatless	2.4	2%
41810610	Plant-based burgers	Chicken, meatless, breaded, fried	2.4	3%
41810610	Plant-based burgers	Chicken, meatless, breaded, fried	2.4	3%
41810600	Plant-based burgers	Chicken, meatless, NFS	2.4	3%
41810600	Plant-based burgers	Chicken, meatless, NFS	2.4	3%
41810400	Plant-based burgers	Breakfast link, pattie, or slice, meatless	2.4	6%
41810400	Plant-based burgers	Breakfast link, pattie, or slice, meatless	2.4	6%
41810250	Plant-based burgers	Bacon bits, meatless	2.4	60%
41810200	Plant-based burgers	Bacon strip, meatless	2.4	15%
41440100	Meal replacement drinks/Meal Replacement Bars	Meal replacement or supplement, liquid, soy-based	2.4	1%
41440050	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Ensure Plus liquid nutrition	2.4	1%
41440010	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Ensure liquid nutrition	2.4	1%
41440010	Meal replacement drinks/Meal Replacement Bars	Meal replacement or supplement, liquid, soy-base, high protein	2.4	1%

41436000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Nutritional supplement for people with diabetes, liquid	2.4	1%
41435710	Meal replacement drinks/Meal Replacement Bars	South Beach Living Meal Replacement Bar	2.4	5%
41435700	Nutritionally fortified snack foods – protein bars, energy bars	South Beach Living High Protein Cereal Bar	2.4	7%
41435120	Nutritionally fortified snack foods – protein bars, energy bars	Zone Perfect Classic Crunch nutrition bar	2.4	5%
41435110	Nutritionally fortified confectionary – hard candies, soft gummies	High protein bar, candy-like, soy and milk base	2.4	4%
41430310	Dry blend powders - nutritionally fortified beverage mixes	Protein diet powder with soy and casein	2.4	6%
41430200	Dry blend powders - nutritionally fortified beverage mixes	Meal replacement or supplement, soy- and milk-base, powder, reconstituted with water	2.4	7%
41430010	Dry blend powders - nutritionally fortified beverage mixes	Protein supplement, powdered	2.4	8%
41430000	Dry blend powders - nutritionally fortified beverage mixes	Protein powder, NFS	2.4	8%
27564570	Plant-based burgers	Frankfurter or hot dog sandwich, meatless, on bread, with meatless chili	2.4	1%
27564560	Plant-based burgers	Frankfurter or hot dog sandwich, meatless, on bun, with meatless chili	2.4	1%
27564430	Plant-based burgers	Frankfurter or hot dog sandwich, meatless, plain, on bread	2.4	2%
27564430	Plant-based burgers	Frankfurter or hot dog sandwich, meatless, plain, on bread	2.4	2%

27564420	Plant-based burgers	Frankfurter or hot dog sandwich, meatless, plain, on bun	2.4	2%
27564420	Plant-based burgers	Frankfurter or hot dog sandwich, meatless, plain, on bun	2.4	2%
11836100	Dry blend powders - nutritionally fortified beverage mixes	Protein supplement, milk-based, Muscle Milk Light, powdered, not reconstituted	2.4	3%
11836000	Dry blend powders - nutritionally fortified beverage mixes	Protein supplement, milk-based, Muscle Milk, powdered, not reconstituted	2.4	3%
11835150	Dry blend powders - nutritionally fortified beverage mixes	Dynatrim, meal replacement, powder	2.4	8%
11835100	Meal replacement drinks/Meal Replacement Bars	Meal replacement, Amway's Nutrilite brand Positrim Drink Mix, powdered nonfat dry milk-based, dry, not reconstituted	2.4	5%
11835000	Meal replacement drinks/Meal Replacement Bars	Meal replacement or nutritional supplement, Cambridge diet formula, powdered, nonfat milk solids base, dry, not reconstituted	2.4	5%
11832000	Meal replacement drinks/Meal Replacement Bars	Meal replacement, protein type, milk- and soy-based, powdered, not reconstituted	2.4	7%
11831500	Dry blend powders - nutritionally fortified beverage mixes	Nutrient supplement, milk-based, high protein, powdered, not reconstituted	2.4	8%
11830990	Dry blend powders - nutritionally fortified beverage mixes	Nutrient supplement, milk-based, powdered, not reconstituted	2.4	22%
11830970	Meal replacement drinks/Meal Replacement Bars	Meal replacement, protein type, milk-based, powdered, not reconstituted	2.4	8%
11830940	Meal replacement drinks/Meal Replacement Bars	Meal replacement, high protein, milk based, fruit juice mixable formula, powdered, not reconstituted	2.4	6%

11830900	Dry blend powders - nutritionally fortified beverage mixes	Protein supplement, milk-based, powdered, not reconstituted	2.4	8%
11830850	Meal replacement drinks/Meal Replacement Bars	High calorie milk beverage, powder, not reconstituted	2.4	8%
11830810	Meal replacement drinks/Meal Replacement Bars	Instant breakfast, powder, sweetened with low calorie sweetener, not reconstituted	2.4	12%
11830800	Meal replacement drinks/Meal Replacement Bars	Instant breakfast, powder, not reconstituted	2.4	7%
11830120	Dry blend powders - nutritionally fortified beverage mixes	Cocoa, whey, and low calorie sweetener, fortified, dry mix, not reconstituted	2.4	12%
11651010	Meal replacement drinks/Meal Replacement Bars	Meal replacement formula, Cambridge diet, reconstituted, all flavors	2.4	1%
11641020	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Meal replacement or supplement, milk based, ready-to-drink	2.4	1%
11641000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Meal supplement or replacement, milk-based, high protein, liquid	2.4	1%
11631000	Meal replacement drinks/Meal Replacement Bars	High calorie beverage, canned or powdered, reconstituted	2.4	1%
11623000	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Meal supplement or replacement, commercially prepared, ready-to-drink	2.4	1%
11622010	Dry blend powders - nutritionally fortified beverage mixes	Diet beverage, powder, reconstituted with skim milk	2.4	48%
11613000	Meal replacement drinks/Meal Replacement Bars	Instant breakfast, powder, sweetened with low calorie sweetener, milk added	2.4	12%

11553120	Ready to drink (RTD) beverages - protein shakes, smoothies etc., sports drinks	Fruit smoothie, with whole fruit and dairy, added protein	2.4	0%
11513855	Dairy alternatives, Dairy analogs	Chocolate milk, made from sugar free syrup with non-dairy milk	2.4	1%
11513855	Dairy alternatives, Dairy analogs	Chocolate milk, made from sugar free syrup with non-dairy milk	2.4	1%
11513805	Dairy alternatives, Dairy analogs	Chocolate milk, made from light syrup with non-dairy milk	2.4	1%
11513750	Dairy alternatives, Dairy analogs	Chocolate milk, made from syrup with non-dairy milk	2.4	1%
11513395	Dairy alternatives, Dairy analogs	Chocolate milk, made from no sugar added dry mix with non-dairy milk (Nesquik)	2.4	1%
11513395	Dairy alternatives, Dairy analogs	Nesquik, chocolate milk, made from no sugar added dry mix with non-dairy milk	2.4	1%
11513385	Dairy alternatives, Dairy analogs	Chocolate milk, made from dry mix with non-dairy milk (Nesquik)	2.4	1%
11513385	Dairy alternatives, Dairy analogs	Nesquik, chocolate milk, made from dry mix with non-dairy milk	2.4	1%
11513375	Dairy alternatives, Dairy analogs	Chocolate milk, made from reduced sugar mix with non-dairy milk	2.4	1%
11513375	Dairy alternatives, Dairy analogs	Chocolate milk, made from reduced sugar mix with non-dairy milk	2.4	1%
11513310	Dairy alternatives, Dairy analogs	Chocolate milk, made from dry mix with non-dairy milk	2.4	1%
11511700	Dairy alternatives, Dairy analogs	Nesquik, chocolate milk, ready to drink, low fat (1%), no sugar added	2.4	1%
11300100	Dairy alternatives, Dairy analogs	Non-dairy milk, NFS	2.4	1%

EXHIBIT 1

Report of the Expert Panel

OPINION OF A GRAS PANEL ON THE SAFETY AND GENERALLY RECOGNIZED AS SAFE (GRAS) STATUS OF FAVA BEAN PROTEIN HYDOLYSATE FOR USE AS AN INGREDIENT IN HUMAN FOOD

Introduction

An independent panel of experts (GRAS Panel), qualified by scientific training and experience to evaluate the safety of food and food ingredients, was requested by Nuritas to determine the safety and Generally Recognized as Safe (GRAS) status of the use of fava bean protein hydrolysate as an ingredient in human food. Fava bean protein hydrolysate is intended for use as a source of protein for enrichment of processed foods. The fava bean protein hydrolysate ingredient is manufactured in accordance with current Good Manufacturing Practice (cGMP) and meets the proposed specifications.

A detailed review based on the existing scientific literature (through July 2022) on the safety of fava bean protein hydrolysate was conducted by ToxStrategies, Inc. (ToxStrategies) and is summarized in the attached dossier. The GRAS Panel members reviewed the dossier prepared by ToxStrategies and other pertinent information and convened on November 9, 2022 via teleconference. Based on an independent, critical evaluation of all of the available information and discussions during a November 9, 2022 teleconference and subsequent reviews/discussions, the GRAS Panel unanimously concluded that the intended uses and use levels described herein for Nuritas' fava bean protein hydrolysate ingredient, meeting appropriate food-grade specifications as described in the supporting dossier (**GRAS Determination of Fava Bean Protein Hydrolysate for Use as an Ingredient in Human Food**) and manufactured according to cGMP, are safe, suitable, and GRAS based on scientific procedures. A summary of the basis for the GRAS Panel's conclusion is provided below.

Summary and Basis for GRAS Determination

Description

Fava bean protein hydrolysate is a protein ingredient derived from fava beans (*Vicia faba* L.) that is 55-65% protein (dry matter basis). The remainder of the ingredient is ash, fat, carbohydrates, and water. The fava bean protein hydrolysate ingredient is also rich in amino acids.

Manufacturing Process

The fava bean protein hydrolysate ingredient (approximately 60% protein) is derived from a fava bean protein starting material in a process that involves a series of steps with common, safe, and suitable processing aids (all have U.S. regulatory approval) and without the use of organic solvents (see Table 4). The protein powder starting material is resuspended in a suitable food grade alkalizing buffer and heated. Following pH adjustment, if required, with an acid suitable for food production, a commercially available, microbial-derived protease enzyme is added to catalyze the hydrolysis reaction.

This enzyme preparation is GRAS in accordance with 21 CFR § 184.1(h)(1), and approved for use in food with no limitation other than current good manufacturing practice. The affirmation of the use this enzyme in food is based upon the following current good manufacturing practice conditions of use: (1) The ingredient is used as an enzyme as defined in § 170.3(o)(9) of this chapter to hydrolyze proteins or polypeptides. (2) The ingredient is used in food at levels not to exceed current good manufacturing practice. After a defined period, the reaction is deactivated with a thermal heat treatment step before spray-drying. These steps ensure that no residual enzyme activity remains in the final product.

Analytical (physical, chemical, and microbiological) results for the fava bean protein hydrolysate product confirm that the finished product meets the proposed specifications as demonstrated by the consistency of production, the lack of impurities and contaminants (e.g., heavy metals, pesticides, aflatoxins, and microbiological contaminants), and is stable for 24 months when stored under recommended storage conditions: in a closed container in a cool (below 100°F) and dry place away from strong light.

History of Use

Fava bean (*Vicia faba* L.), also referred to as broad bean, horse bean, faba bean, and field bean, is an early legume crop and belongs to the *Fabaceae* family. The fava bean has a long history of use as both animal feed and food, likely because of its valuable content of both protein and energy (Crepon et al., 2010). In the human diet, it is mostly the seed grain that is consumed, while the pods are used as feed. The fava bean is a rich source of fiber and non-nutrient secondary metabolites that have been shown to be beneficial to human health. The protein content of the fava bean is about twice that of cereals and several times that of root tubers. Fava bean seeds are a rich source of proteins, carbohydrates, fiber, vitamins, and minerals (Aune et al. 2011).

Protein is found throughout the body - in bone, muscle, hair, skin, and virtually every tissue and body part. At least 10,000 different proteins exist in the human body. Proteins are made up of amino acid building blocks, some of which must be provided by the diet (essential amino acids) as they cannot be made by the body. While animal sources of protein tend to deliver all the essential amino acids, plant protein sources also deliver the majority of the required amino acids and have become an important option for added protein in processed foods. Plant and cereal grain sources of added protein currently used in food include peas, soy, lentils, chickpeas, beans, canola, rice, potatoes, and wheat.

Humans have consumed fava beans, as well as proteins from these sources for centuries, along with proteins from many food sources such as meats, dairy, fruits, vegetables, nuts, and seeds. The USDA recommends that half of a person's meal consist of grains and protein foods (equal amounts), and the other half should contain fruits and vegetables, with a serving of dairy. The Institute of Medicine (IOM, 2005) recommends that adults consume a minimum of 0.8 g protein/kg and has set a range for acceptable protein intake of 10 - 35% of daily calories. In the US, the recommended daily allowance (RDA) of

protein is 56 and 46 grams/day for adult men and women (>19 years of age), respectively. Physically active persons on normal diets may easily exceed this level, and individuals involved in bodybuilding ingest much higher levels of protein (WHO, 2002). WHO (2002) recommends body weight-based protein consumption rates for both genders. For example, the safe protein consumption level for a 40-kg adult is 33 g/day, and that for an 80-kg adult is 66 g/day. For adults, the protein requirement per kg body weight is considered to be the same for both sexes, at all ages, and for all body weights within the acceptable range. The value accepted as a safe level of intake is 0.83 g/kg per day, for proteins with a protein digestibility-corrected amino acid score value of 1.0 (WHO, 2002). While WHO states that no safe upper limit has been identified, they also indicate that it is unlikely that intakes of twice the safe level are associated with any risk.

Many protein products are currently available in the marketplace. To date, FDA has reviewed extensive published information and data as part of GRAS notifications for animal and plant-based protein isolates and concentrates and subsequently issued “no questions letters”.

Intended Use and Intake Assessment

The focus of this GRAS determination of fava bean protein hydrolysate is for its use in certain specified conventional foods at levels similar to those that have been recognized in previous GRNs for current plant-based protein sources such as soy (GRN No. 134; FDA 2004), canola (GRN Nos. 327, 386, 683; FDA 2010, 2011, 2017a), pea (GRN Nos. 182, 581, 608, 788; FDA 2005, 2015b, 2016a, 2018b), wheat (GRN Nos. 26 and 182; FDA 1999, 2005), rice (GRN No. 609, 944; FDA 2016b, 2020b), whey (GRN Nos. 37, 633; FDA 2000, 2016c), potato (GRN No. 447; FDA 2013), oat (GRN 575; FDA 2015a), mung bean (GRN 684; FDA 2017b), hemp seed (GRN 771; FDA 2018a) fava bean protein (GRN 879; FDA 2020a) and rice protein hydrolysate (GRN 944: 2020b). Similarly, fava bean-derived protein hydrolysate is intended for use as a source of protein for enrichment of certain specified processed foods.

The proposed uses of the fava bean protein hydrolysate ingredient for enrichment of foods include the following food categories (see Table below):

- Dry blend powders - nutritionally fortified beverage mixes
- Meal replacement drinks/meal replacement bars
- Dairy alternatives, dairy analogs
- Plant-based burgers
- Ready to drink (RTD) beverages - protein shakes, smoothies, sports drinks, etc.
- Nutritionally fortified confectionary – hard candies, soft gummies etc.
- Nutritionally fortified snack foods – protein bars, energy bars etc.
- Baked goods – nutritionally fortified cookies, biscuits, cereal bars
- Meat alternatives-such as vegan burger, etc.

The fava bean protein hydrolysate ingredient will be added at a maximum of 2.4 g/serving or given a protein content of approximately 60%, approximately 1.44 g protein/serving.

As described in numerous GRAS Notifications, the typical uses of protein for enrichment of foods include bakery products, snack foods, nutritional beverages such as high-protein drinks and milkshakes, instant powdered nutritional beverages, vegetarian food products and meat analogues, dairy products, and meal replacements/nutrition bars.

As the previously submitted GRNs on plant-derived proteins have demonstrated, the proposed use concentrations and the variety of food uses containing these plant-derived proteins, combined with the large average daily consumption of the described foods, results in a calculated daily intake of the added protein from these sources being a substantial fraction of the Recommended Dietary Allowance (RDA; 46 g/day for women over 19 years of age and 56 g/day for men over 19 years of age), and can even exceed it at the 90th percentile consumption. This was the case for GRN No. 327 (cruciferin-rich canola/rapeseed protein isolate and napin-rich protein canola/rapeseed protein isolate).

Nuritas' proposed fava bean protein hydrolysate is intended only to be an alternative source of protein for use in certain specified foods identified in this dossier. As was concluded in numerous other GRAS notifications for plant-based dietary proteins for which FDA issued "no objection" letters, it is not realistic to expect that the actual consumption of foods containing fava bean protein hydrolysate would result in daily protein consumption greater than the existing RDA for protein. It is reasonable to expect that most of the population's daily intake of protein will remain in the form of unprocessed foods, including meat, poultry, fish, and legumes (FDA, 2010, 2011).

In summary, the proposed uses of the fava bean-derived protein ingredient will not result in an increase in the overall consumption of protein, but simply provide a relatively minor proportion of well-characterized, alternative protein from fava beans for use in certain specified conventional foods.

Safety Data

Fava beans and other legumes, and the protein from these sources, have been commonly consumed as food and feed around the world for decades (Crepon et al., 2010; Schmandke et al., 2010). Humans have consumed fava beans for centuries, along with proteins from many food sources such as meats, dairy, fruits, vegetables, nuts and seeds. Many protein products are currently available in the marketplace. To date, FDA has reviewed extensive published information and data as part of GRAS notifications for animal and plant-based protein isolates and concentrates and subsequently issued "no questions letters"

The fava bean is consumed in both raw and processed forms. In the human diet, it is mostly the seed grain that is consumed, while the pods are used as animal feed. The pods provide macro-, micro- and non-nutrient phytochemicals and could be used as a source of

functional compounds. Fava beans contain approximately 250 g protein/kg seed and provides 320 kcal/100 g dry weight of energy. Fava bean, its fractions, and its processing products (grains, hulls, and flours) also contain anti-nutritional factors, but soaking and cooking are able to reduce the amounts of the anti-nutritional factors up to 100%, thus limiting any safety concerns (Jamalian et al., 1999).

Fava bean is also a rich source of fiber and non-nutrient secondary metabolites that have been shown to be beneficial to human health (Aune et al. 2011). The protein content of fava beans is about twice that of other cereal crops and is several times that of root tubers (Wu Leung et al., 1968). However, the biological value of untreated fava beans is negatively affected by the presence of anti-nutritional factors such as the favism-inducing factors (aglycones: divicine and isouramil) produced from metabolism of vicine and convicine (Jamalian et al., 1999; Frank, 2005; Luzzatto and Arese, 2018). The ingestion of untreated fava beans is associated with the hemolytic disease “favism” in certain glucose-6-phosphate dehydrogenase-deficient individuals (Kavehmansh et al., 2016; Skold et al., 2017; Luzzatto and Arese, 2018; Al-Dubai et al. 2021; Xin et al., 2022; Diegues et al., 2022).

The toxicity of vicine, convicine, divicine, and isouramil are discussed in the safety section below, and the concentrations of the anti-nutrients vicine and convicine in three lots of the fava bean protein hydrolysate ingredient are presented below.

Alkaloid concentration in three lots of fava bean protein hydrolysate (mg/g)*

Alkaloid	Batch No. sh_CU6G3M	Batch No. sh_YO75MA	Batch No. sh_FDE4MQ
Vicine	14.1	11.3	0.54
Convicine	4.7	4.1	0.155
Sum	18.8	15.4	0.695

Levodopa, also known as L-DOPA or L-3,4-dihydroxyphenylalanine, is a chemical precursor to dopamine and a common treatment for Parkinson’s disease. L-DOPA naturally occurs in fava beans, but its levels decrease significantly in heated/cooked beans (Carador-Martinez et al., 2012; Multari et al, 2015). The level of L-DOPA in the fava bean protein hydrolysate was determined to be approximately 0.14% or 1400 mg/kg. If a person was to consume the 90th percentile (users only) intake of 12.4 g/day, the maximum calculated exposure is approximately 17 mg of levodopa, which is approximately 5.7% of the typical recommended daily dose (300 mg) and approximately 1.1% of the maximum recommended daily dose (1500 mg) for the treatment of Parkinson’s disease.

Given the long history of global human consumption of fava beans as food (and the protein contained therein), the safety of the fava bean protein hydrolysate ingredient derived from them is supported by the historical consumption and general lack of toxicity. As would be expected for a food that has been consumed by humans for

centuries, fava beans and fava bean proteins have not been subjected to traditional toxicology studies. Given the publicly available information and data on the safety of fava beans and fava bean proteins, the conduct of toxicity studies was considered unnecessary and not an ethical use of animals.

It has been long recognized that fava beans contain “anti-nutritional” compounds. Those that are of most concern are vicine (2,6-diamino-4,5-dihydroxypyrimidine 5-(beta-D-glucopyranoside) and convicine (2,4,5-trihydroxy-6-aminopyrimidine 5-beta-D-glucopyranoside), both of which are glycosidic aminopyrimidine derivatives. The vicine and convicine content of raw fava beans have been reported to range from 0.02% to 1.46% (dry weight basis) (Khamassi et al., 2013). Vicine and convicine are hydrolyzed in the fresh fava bean by β -glucosidase to form the aglycones divicine and isouramil (Figure 2, below). As reviewed in GRN 879, these aglycones are the compounds responsible for “favism,” by causing the oxidation of glutathione in red blood cells that cannot be sufficiently reversed in affected individuals who are mostly males. The erythrocytes become rigid, experience aggregation of their proteins, produce methemoglobin and have their enzymes inactivated due to oxidative damage that cannot be controlled or reversed. The damaged red blood cells are then removed by macrophages in the spleen and liver, which could lead to a potentially fatal hemolytic anemia.

For this reason, potential adverse reactions to fava bean ingestion are limited to a small, but significant, number of males with this X-linked genetic disease. There are many genetic variants of G6PD deficiency, with one common variant observed primarily in Africans and African-Americans and another in people with Mediterranean origins (e.g., Greeks, Italians, Armenians, several Semitic populations and others). Some Asian and middle eastern populations are also affected. G6PD deficiency in the U.S. is observed in approximately 10% of African-American males (Frank, 2005).

In the European, Middle Eastern, African, and Chinese areas where individuals prone to favism live, it is estimated that one meal can result in an intake of 2000 mg of vicine and 1000 mg of convicine (WHO, 1992). Numerous cases of favism related to consumption of raw, unprocessed fava beans by G6PD-deficient individuals have been reported (Alkhaaldi et al., 2014; Eghbal et al., 2012; Kavehmansh et al., 2016; Skold et al., 2017; Luzzatto and Arese, 2018; Al-Dubai et al. 2021; Xin et al., 2022).

Arese et al. (2007) reported levels of vicine and convicine in FEVITA (a low vicine/convicine protein ingredient) were 10 – 20x lower than traditional cultivars (i.e., 6-14 mg/g). Arese et al. (2009) studied seven hemizygous males with G6PD deficiency using the FEVITA fava bean, which is low in vicine and convicine. The subjects were fed 450 g of the FEVITA fava bean which contained 135 - 315 mg of vicine and convicine. Given the worst-case vicine and convicine level of 18.8 mg/g in the proposed ingredient, times the mean and 90th percentile (users only) intakes of 6.4 and 12.4 g/day, respectively, results in an intake of approximately 120 mg and 233 mg vicine and convicine in the diet, amounts that are in the range of dose levels employed in the Arese et al. (2009) study in which no hemolytic effects were reported.

The potential for fava bean proteins to cause an immune response is rare but consistent with similar, known allergic responses to other legumes. Although fava beans are not listed as one of eight major allergen groups by the FDA under the Food Allergen Labeling and Consumer Protection Act of 2004 (Public Law 108-282, Title II), the fact that the allergenicity of fava bean protein has been shown clinically (Mur Gimeno et al., 2007; Kumar et al., 2014; Damiani et al., 2011; Jensen et al., 2008) clearly indicates that labeling of the presence of fava bean protein in any food product is warranted and recommended. More importantly, given the incidence of G6PD deficiency in the US, consumers may not expect fava bean protein to be present in their food. The Panel recommends that the ingredient labeling for the fava bean protein hydrolysate product clearly state that it contains “fava bean protein” and that individuals who wish to avoid fava bean or fava bean protein consumption for any reason would be able to identify the presence of a fava bean–derived ingredient.

Taken together, the available published safety data demonstrate that the fava bean protein hydrolysate from *Vicia faba* has little potential for toxicity when used in foods for human consumption.

General Recognition of the Safety of Fava Bean Protein Hydrolysate

The intended use of the fava bean protein hydrolysate ingredient in human food has been determined to be safe through scientific procedures set forth in 21 CFR§170.3(b), thus satisfying the so-called “technical” element of the GRAS determination, based on the following:

- Fava bean protein hydrolysate is manufactured from commercially available fava beans, following current cGMP for food (21 CFR § Part 110). The raw materials and processing aids used in the manufacturing process are food grade and/or approved for use in food. The fava bean protein hydrolysate product has been characterized appropriately, contains approximately 60% protein, and meets appropriate food-grade specifications.
- Fava beans have been consumed as food (and the protein contained therein) for centuries, along with many other food sources of protein (e.g., meats, dairy, fruits, vegetables, nuts).
- The biological value of untreated fava beans is negatively affected by the presence of anti-nutritional factors such as the favism-inducing factors (the aglycones: divicine and isouramil produced from metabolism of vicine and convicine). The ingestion of untreated fava beans is associated with an occurrence of the hemolytic disease “favism” in certain glucose-6-phosphate dehydrogenase-deficient humans.
- These anti-nutrients represent a known safety concern. Calculated daily intakes of the fava bean protein hydrolysate of approximately 120 mg and 233 mg vicine and convicine in the diet (mean and 90th percentile), represents amounts of vicine

and convicine that are in the range of dose levels employed in the Arese et al. (2009) study in which no hemolytic effects were reported.

- The proposed uses of the fava bean protein hydrolysate ingredient in specified foods will provide an alternative to other dietary sources of protein and will not increase the overall consumption of protein but will simply provide another source of well-characterized protein for use in food products.
- Fava beans are a rich source of fiber and non-nutrient secondary metabolites shown to be beneficial to human health. The protein content is about twice that of other cereals and several times that of root tubers.
- FDA has reviewed published information and data on many protein products as part of GRAS notifications for animal and plant-based protein isolates and concentrates and subsequently issued “no objection letters.” Examples include GRN No. 26 (isolated wheat protein); GRN No. 37 (whey protein isolate and dairy product solids); GRN No. 168 (poultry protein); GRN No. 182 (hydrolyzed wheat gluten isolate; pea protein isolate); GRN No. 313 (beef protein); GRN No. 314 (pork protein); GRN 386 (canola protein isolate and hydrolyzed canola protein isolate); GRN No. 447 (potato protein isolates); GRN No. 575 (oat protein); GRNs No. 58, 608, and 788 (pea protein); GRN 879 (fava bean protein); and GRN 944 (rice protein hydrolysate).
- Given the long history of global human consumption of fava beans as food (and the protein contained therein), the safety of the fava bean protein hydrolysate ingredient derived from them is supported by a history of consumption and general lack of toxicity. As would be expected for a food that has been consumed by humans for centuries, fava beans and fava bean proteins have not been subjected to traditional toxicology studies. However, the available preclinical and clinical study data support its safe use as proposed.
- Concerns related to the allergenicity of fava bean protein as well as the effect of fava beans consumption by G6PD deficient individuals will be addressed through appropriate labeling of food products as containing fava bean protein and individuals who wish to avoid fava bean protein consumption would be able to identify the presence of a fava bean–derived ingredient.
- The body of publicly available scientific literature on the consumption and safety of fava beans and fava bean protein is sufficient to support the safety and GRAS status of the proposed fava bean protein hydrolysate ingredient.

Conclusions of the Expert Panel

We, the undersigned independent, qualified members of the GRAS Panel, have individually and collectively critically reviewed the published and ancillary information pertinent to the identification, use, and safety of Nuritas' fava bean protein hydrolysate ingredient for use as an alternative source of protein in specified foods. We unanimously conclude that the intended use of Nuritas' fava bean protein hydrolysate ingredient, produced consistent with current good manufacturing practice (cGMP) and meeting the appropriate food-grade specifications, as presented in the supporting dossier "GRAS Determination of Fava Bean Protein Hydrolysate for Use as an Ingredient in Human Food", is safe.

We, the members of the GRAS Panel, further unanimously conclude that the intended uses and use levels of Nuritas' fava bean protein hydrolysate ingredient in specified foods, produced consistent with current good manufacturing practice (cGMP) and meeting the appropriate food-grade specifications as presented in the supporting dossier is Generally Recognized as Safe (GRAS) based on scientific procedures under the conditions of intended use in foods as described herein.

It is our professional opinion that other qualified experts critically evaluating the same information would concur with this conclusion.

Paul Damian, PhD, MPH, DABT
Consultant
Damian Applied Toxicology, LLC

Date

Carol A. Knight, PhD
Consultant
Knight International

Date

Stanley M. Tarka, Jr., Ph.D., Fellow, ATS
The Tarka Group, Inc.
The Pennsylvania State University, College of Medicine

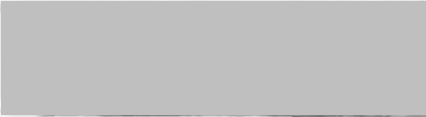
Date

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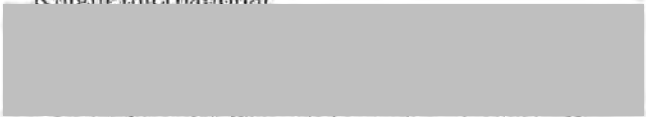
It is our professional opinion that other qualified experts critically evaluating the same information would concur with this conclusion.


Paul Damian, PhD, MPH, DABT
Consultant
Damian Applied Toxicology, LLC

September 26, 2023
Date


Carol A. Knight, PhD
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25 September 2023
Date


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From: [Don Schmitt](#)
To: [Hall, Karen](#)
Cc: Shah.Shweta@Nuritas.com; [Alexander East](#); [Amy Sheppard](#)
Subject: [EXTERNAL] Re: Regarding GRN 1166
Date: Wednesday, September 25, 2024 12:19:34 PM
Attachments: [image013.png](#)
[image014.png](#)
[image015.png](#)
[image016.png](#)
[image017.png](#)
[image018.png](#)
[image001.png](#)
[image002.png](#)

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Hi Karen,

We agree with the new Table 8 as discussed on our call and included in your email.

We agree with statements in your email: grits and oatmeal removed, dietetic biscuits only, only hard candies and gummies with high or added vitamin C, no cereal bars, meatless chicken included.

We hope this resolves the issue.

Best regards,

Don

Donald F. Schmitt, M.P.H.
Senior Managing Scientist



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From: Hall, Karen <Karen.Hall@fda.hhs.gov>
Date: Wednesday, September 25, 2024 at 8:26 AM
To: Don Schmitt <dschmitt@toxstrategies.com>
Subject: Regarding GRN 1166

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Good Morning Don,

Thank you for your flexibility on the September 10th meeting. We have formulated our question based on what was discussed during the meeting.

1. In the amendment dated July 10, 2024, you provided a revised Table 8 including the intended food uses of fava bean protein hydrolysate. After comparing Table 8 and the food codes used in the dietary exposure estimate (Appendix A) we have several comments regarding your intended uses:
 - Grits and oatmeal
Only the food codes for grits and oatmeal prepared with non-dairy milk were included in the dietary exposure estimate indicating that the intended use is in a non-dairy milk component of the prepared grits and oatmeal. As such, grits and oatmeal should be removed from the intended uses (the use in the non-dairy milk component is covered by the “Milk analogs” category).
 - Cookies, biscuits
Food codes for dietetic or high- fiber cookies were the only food codes used in your dietary exposure estimate. If the intended use is only in these types of cookies, this should be specified in Table 8. Furthermore, for biscuits listed as a specific food subcategory, you only included one food code in your dietary exposure estimates. It is not clear whether the intended use is in all biscuits or only in dietetic biscuits. The intended use in biscuits should be clarified.
 - Hard candies, gummies
The food codes used in the dietary exposure estimate represent only hard candies and gummies with high or added vitamin C. If the intended use is only in these types of candies and gummies, this should be specified in Table 8.
 - Cereal bars
There are no food codes representing cereal bars in the dietary

exposure estimate. Therefore, cereal bars should not be listed in Table 8. Please clarify if the ingredient is intended for use in cereal bars.

- Meatless pork/beef

In addition to the food codes for meatless pork and beef, the food codes for meatless chicken were used in your dietary exposure estimate. If you intend to use the ingredient in meatless chicken, this use should be listed in Table 8.

Based on the above, we suggest that you describe the intended uses of fava bean protein hydrolysate as presented in Table 8, below. Please confirm that you concur with this description of the intended uses or provide revisions, if needed.

Table 8. Food categories and maximum use levels for fava bean protein hydrolysate

Food categories	Maximum use level (%)
Protein powders	30
Instant breakfast powders	12
Sport drink powders	7
Meal replacement drinks	1
Meal replacement bars	5
Milk analogs	1
Ready-to-drink (RTD) protein shakes	1
RTD sports drinks	1
Hard candies with vitamin C	12
Gummies with vitamin C	12
Protein bars	7
Energy bars	7
Cookies, dietetic or high fiber	7
Biscuits	16
Veggie burgers	6
Meatless fish sticks	2
Meatless pork/beef/chicken	2

Please respond within 2 business days.

Kind Regards,
Karen

Karen M. Hall (she/her/hers)

Regulatory Review Scientist

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